

SECTION – A

There are **FOUR** questions in this section. You **must** answer the **first** question and any **TWO** out of the other **three** questions.

1. (a) Find the z-transform and the associated region of convergence for the following discrete-time function (10)

$$x[n] = a^n u[n]$$

Here $u[n]$ is the discrete-time unit step function.

- (b) Demonstrate whether the s -domain coordinate $(-a, 0)$ included in the region of convergence of the Laplace transform of $x(t)$, where $x(t)$ is a time domain signal defined as (10)

$$x(t) = \begin{cases} e^{-at}, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

- (c) Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then analyze how $x(t)$ is uniquely determined by its samples (15)

$$x(nT), n=0, \pm 1, \pm 2, \dots, \text{ if } \omega_s > 2\omega_M, \text{ where } \omega = \frac{2\pi}{T}.$$

2. (a) State and prove the time- and s -domain differentiation properties of the Laplace transform. (10)

- (b) Determine the time-domain signal from the following Laplace transform: (10)

$$X(s) = \frac{s+2}{s^2+7s+12}, \quad -4 < \operatorname{Re}\{s\} < -3$$

- (c) Determine the Laplace transform and the associated region of convergence for the following function (15)

$$x(t) = e^{-b|t-2|}$$

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3. (a) A discrete-time signal $x[n]$ is defined as (10)

$$x[n] = \begin{cases} a^n, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

Show that the region of convergence of the z-transform of $x[n]$ will not include the origin.

- (b) Derive the contour integral formula of the inverse z-transform. (10)

- (c) Determine the discrete-time signal form the following z-transform: (15)

$$X(z) = \log(1 + az^{-1}), \quad |z| > |a|$$

4. (a) For any general continuous-time signal $x(t)$, establish a relation between its Laplace and Fourier transform. (10)

- (b) Consider a sequence $x[n]$ whose Fourier transform $X(j\omega)$ has the property

$$X(j\omega) = 0 \quad \text{for} \quad \frac{2\pi}{9} \leq |\omega| \leq \pi.$$

Determine the lowest rate at which $x[n]$ may be sampled for its perfect reconstruction. (10)

- (c) Compare the computational complexities of the discrete Fourier transform and the Radix-2 fast Fourier transform. (15)

SECTION - B

There are **FOUR** questions in this section. **You must answer the question no 5, which contains 45 marks.** Answer any **TWO** questions from 6 - 8, each of which contains **30** marks in total.

5. (a) A signal $f(t)$ is defined as follows:

$$f(t) = \begin{cases} t+1 & -2 \leq t \leq 0 \\ -\frac{t}{2} + 1 & 0 < t \leq 2 \\ 0 & \text{everywhere else} \end{cases}$$

Answer the questions i, ii and iii with respect to $f(t)$.

- i. Draw the graph of $f(t)$. (3)

- ii. Determine the even-odd decomposition of $f(t)$. Apply the definitions of both the even function and the odd function in the decomposition of $f(t)$. (6)

- iii. Draw the graphs of the even function and odd function in the decomposition of $f(t)$. (6)

- (b) A function $X(j\omega)$ is defined as follows:

$$X(j\omega) = \begin{cases} 1 & \text{if } |\omega| < 3\pi \\ 0 & \text{otherwise} \end{cases}$$

The magnitude and angle of the Fourier Transform $X_1(j\omega)$ of a signal $x_1(t)$ are shown in Figure 1.

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Contd ... Q. No. 5

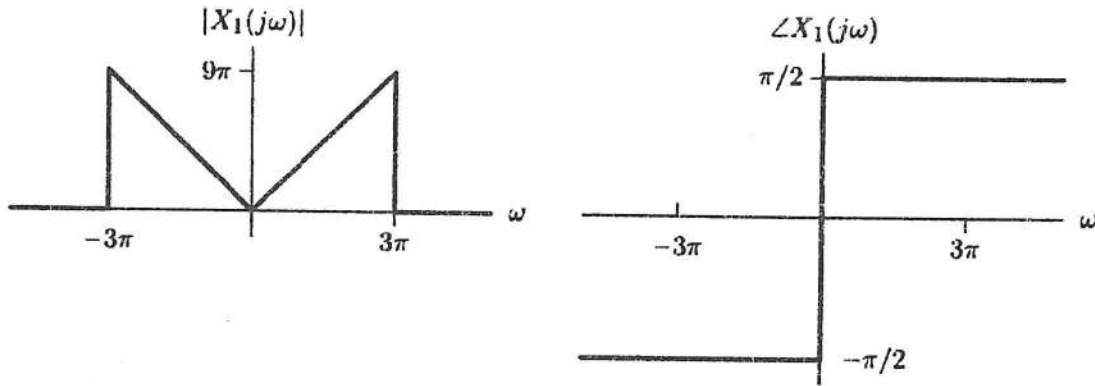


Figure 1: The magnitude and angle of $X_1(j\omega)$ for question 5(b)

Answer the questions i, ii, iii, iv and v. You need not to draw any graphs, but you need to analyze accordingly.

- Write down the definition of $X_1(j\omega)$. (2)
 - Determine the Inverse Fourier Transform $x(t)$ of $X(j\omega)$. (8)
 - Express $X_1(j\omega)$ in terms of $X(j\omega)$. (2)
 - Express $x_1(t)$ in terms of $x(t)$ by using the properties of Fourier Transform. (3)
 - Determine $x_1(t)$. (5)
- (c) Is the signal $x[n] = 2\cos\left(\frac{16\pi n}{63} + \frac{\pi}{3}\right)$ a periodic signal? Explain your answer. (5)
- (d) Determine whether the signal $x(t)$ is real-valued and even. Here $x(t)$ is a continuous-time periodic signal presented in its Fourier series representation. (5)

$$x(t) = \sum_{k=0}^{100} \left(\frac{1}{2}\right)^k e^{jk\frac{2\pi}{50}t}$$

6. Suppose, you have two signals $f(t)$ and $\text{sign}(t)$ defined as follows:

$$f(t) = \begin{cases} 0 & t < 0 \\ e^{-at} & t > 0, \text{ here } a > 0 \end{cases}$$

$$\text{sign}(t) = \begin{cases} -1 & t < 0 \\ 1 & t > 0 \end{cases}$$

Answer the following questions. You need not to draw any graphs.

- Define $f(-t)$. (3)
- Determine the Fourier Transforms of $f(t)$ and $f(-t)$. (12)
- Define $\text{sign}(t)$ in terms of $f(t)$ and/or $f(-t)$. (5)
- Determine the Fourier Transform of $\text{sign}(t)$ by making use of the properties of Fourier Transform and the results obtained in (b). Explicitly mention the Fourier Transform properties you used. (10)

Contd P/4

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7. (a) The graph of a signal $x(t)$ is shown in Figure 2. Assume that $x(t)$ equals 0 when $t < -2$ or $t > 2$.

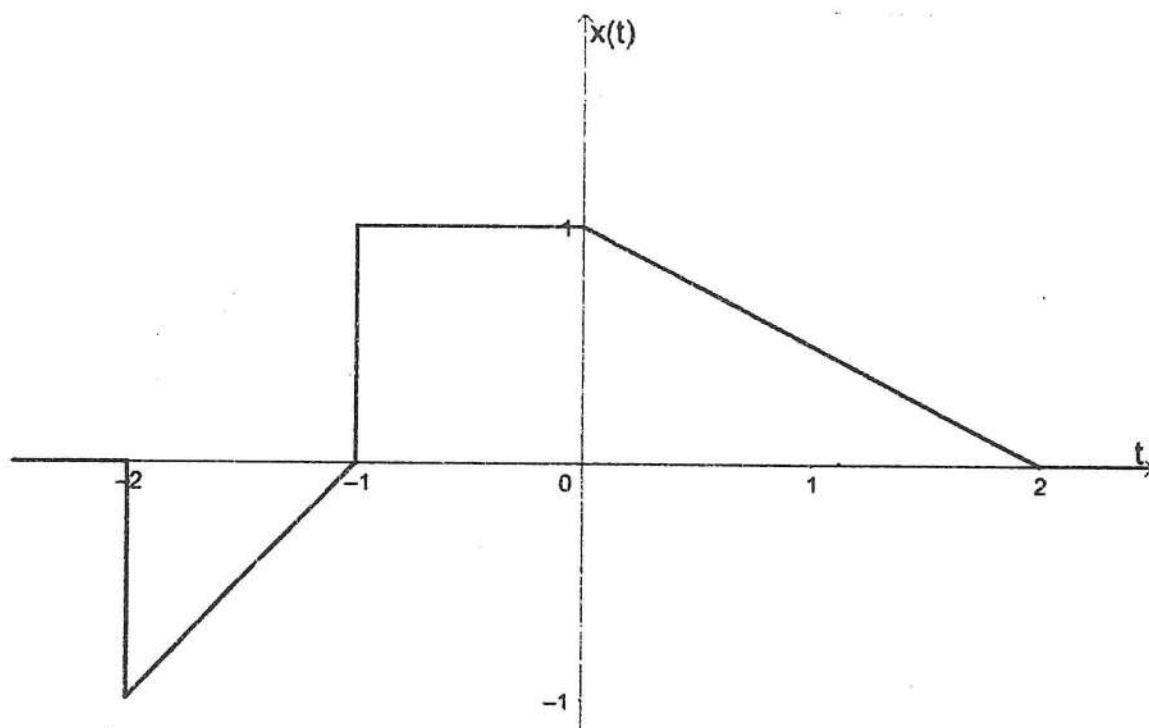


Figure 2: The graph of $x(t)$ for question 7

Answer the questions (a), (b) and (c) with respect Figure 2.

- (a) Write down the piece-wise definition of $x(t)$. (3)

- (b) Determine the signal $x(-2t+2)$ by making use of the properties of signals. Explicitly mention the properties you used in your answer. Finally, draw the graph of the signal. (10)

- (c) Determine the signal $x\left(-\frac{t}{2}+1\right)$ by making use of the properties of signals. Explicitly mention the properties you used in your answer. Finally, draw the graph of the signal. (10)

- (d) Discrete signals have many properties. Suppose, you have an image that you are using as an example of a discrete signal. Now, you need to perform the following tasks on this signal. Mention the specific signal properties that could be useful in accomplishing each task. You just need to mention the specific properties, no need to write details. (7)

- Image Compression
- Blurring the image
- Sharpening the image
- Flipping the image
- Zooming in or out of the image
- Determining the image contrast
- Detecting edges in the image

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8. (a) Compute the convolution $y[n] = x[n] \times h[n]$ for the following signal pairs. (20)

$$x[n] = 3^n u[-n-1] + \left(\frac{1}{3}\right)^n u[n]$$

$$h[n] = \left(\frac{1}{4}\right)^n u[n+3]$$

- (b) Determine whether each of the following systems satisfies these two properties: causality and stability (10)

- $h_1(t) = e^{2t} u(-2-t)$
- $h_2(t) = te^{-t} u(t)$

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2022-2023

Sub: **CSE 311** (Data Communication)

Full Marks: 210

Time: 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – AThere are **FOUR** questions in this section. Answer to **Question no. 1 (one)** is **Compulsory**.Answer any **TWO** questions from Questions 2-4.

1. (a) Demonstrate that the coefficients (D_n) of the exponential Fourier series of an even symmetric periodic signal are real whereas those of an odd symmetric periodic signal are imaginary. Now justify the statement for the signal $\cos 2\pi f_0 t$. The symbols carry their usual meanings. (17)

(b) Derive and draw the Fourier spectrum of the standard rectangular signal $\Pi\left(\frac{t}{2B}\right)$.

Comment on the nature of the frequency spectrum of practical signals. Now analyze why and how sampling process changes the frequency band of a practical signal. The symbols carry their usual meanings. (18)

2. (a) The filtering of a signal $G(f)$ by a filter $H(f)$ in frequency domain results in $G(f)H(f)$. Derive its time domain equivalent using necessary transformation. Prove that the same output is obtained when the corresponding time domain signal $g(t) = \mathcal{F}^{-1}(G(f))$ is passed through a linear time invariant system with impulse response $h(t) = \mathcal{F}^{-1}(H(f))$. Use only time domain analysis for the later proof. The symbols carry their usual meanings. (20)

(b) Explain the differences between a distortionless system and an all-pass system. How will you determine that a system is distortionless? (15)

3. (a) Explain the necessity of different categories of bit stuffing in digital communication. Explain the DM1/2 multiplexer format shown in Figure for 3(a) and describe how positive bit stuffing is done in DM1/2 multiplexer. Identify the **exact bit position** if data of channel B and D are to be stuffed. (15)

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

Figure for 3(a)

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Contd ... Q. No. 6

- (b) A multiple-tone amplitude-modulated (AM) signal is generated by modulating a sinusoidal carrier signal with three sinusoidal baseband signals. The percent modulation index is 58%, 78% and 38%, respectively, for the three baseband signals. Can you identify any problem with this AM transmission? Consider that the carrier signal has a peak amplitude of 5 volts. If there indeed is some problem with the transmission, then state the reason behind it and propose a new peak amplitude for the second baseband signal with a 78% modulation index so that the problem gets resolved. (10)
- (c) Show that for 100% amplitude modulation, the total power transmitted by an antenna is 1.5 times the carrier power, and each sideband contains around 16.67% of the total transmitted power. (10)
7. (a) Derive the expression of instantaneous frequency from the expression of phase angle for a phase-modulated (PM) signal. Likewise, derive the expression of phase angle from the expression of instantaneous frequency for a frequency-modulated (FM) signal. Also, describe how we can use a phase modulator to generate an FM signal and a frequency modulator to generate a PM signal, with appropriate diagrams. (15)
- (b) A radio station transmits a frequency-modulated (FM) audio signal, where the modulating (baseband) audio signal has a 980 Hz frequency and a 3 volts peak amplitude. The frequency deviation in the FM audio signal is 5 kHz . Again, the radio station transmits a different FM audio signal, where the modulating audio signal has a 1150 Hz frequency and a 4.5 volts peak amplitude. Compute the modulation index as well as the bandwidth of the FM audio signal using Carson's rule in both cases. Also, explain the motivation behind using Carson's rule in this scenario to calculate bandwidth. (15)
8. (a) Briefly discuss how amplitude shift-keying (ASK) and phase shift-keying (PSK) can be implemented to modulate a sinusoidal carrier signal with a digital baseband signal in digital signal modulation. (5)
- (b) Consider a piece of binary data represented by a bit sequence of 101101001011. Draw the waveform of this binary data when the corresponding digital signal is generated using (i) Manchester encoding (IEEE802.3 convention) and (ii) differential Manchester encoding schemes. (10)
- (c) The musical alphabet is a system of seven letters that represent the seven tones in a scale. This alphabet consists of seven English characters: A, B, C, D, E, F and G. Design a simple encoding scheme for converting these characters into bit strings. Also develop a line-coding scheme to generate digital signals from strings consisting of these musical alphabets. Show the waveform for a digital signal generated from a string FEDE using your proposed scheme. (15)

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L-3/T-1 B. Sc. Engineering Examinations 2021-2022

Sub: **CSE 311** (Data Communication)

Full Marks: 210

Time: 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks

SECTION – AThere are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) Derive the Fourier spectrum of a gate signal of width B . Also, find the exponential Fourier series of a periodic gate signal of the same width and period $2B$. Draw all four time domain and spectral domain representations. Comment on and correlate your findings in the spectral representations. (20)
(b) T1 carrier system multiplexes 24 voice signals to produce Digital Signal Level 1 (DS1) which is then transmitted. How are these individual signals separated at the receiving end? (15)
2. (a) Differentiate between all-pass and distortionless transmission systems. Draw the frequency response of a distortionless transmission system. Justify from its time domain representation why it is unrealizable. (20)
(b) Draw the graphs of the NRZ-I and 2B1Q schemes for the following two data streams. Assume that the last signal level has been positive. (15)
Data streams: 0110011001100110 and 1100110011001100
3. (a) With appropriate examples and mathematical derivations, establish the relationship between time compression and frequency compression of a signal. Nonlinear quantization uses logarithmic compression of a signal, yet it does not change the bandwidth of the signal. Why is it so? Why is nonlinear quantization necessary? (20)
(b) While sampling a signal $g(t)$ band limited to B at Nyquist sampling rate, all samples are found to be zero except at time $t = 0$, where it is A . Derive the most simplified expression of the signal $g(t)$ from its sample values. Justify and correlate the expression you derived and the sample values you used. (15)

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4. (a) In a video surveillance system, data from 24 video cameras are brought to a processing center and are sampled, quantized, binary coded and time-division multiplexed. The multiplexed data are then transmitted to a monitoring system over a cable. Assume the bandwidth of all video signals is 200 Hz. The maximum acceptable error in sample amplitudes is 0.25% of the peak signal amplitude. The sampling rate is exactly the Nyquist rate. Derive the minimum cable bandwidth needed to transmit these data. (18)

(b) "Delta modulation is a special case of differential pulse coded modulation. However, they differ on what types of signals they can process, e.g., continuous time and discrete time signals." Justify the statement with appropriate mathematical derivations and block diagrams of both modulation techniques. (17)

SECTION - B

There are FOUR questions in this section. Answer any THREE questions.

5.

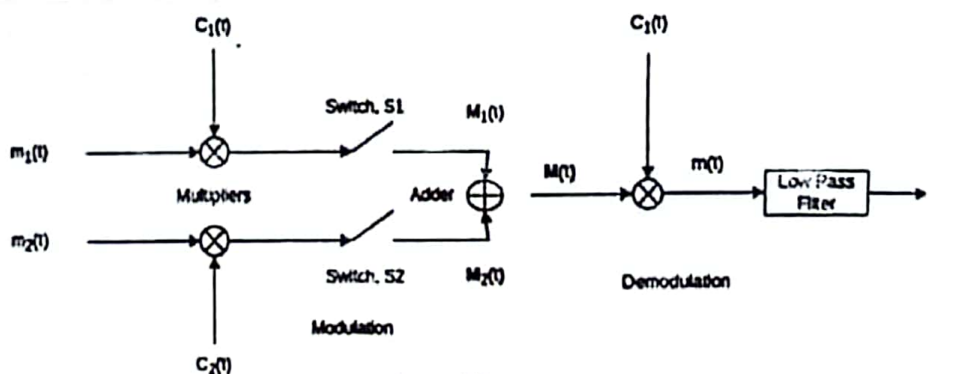


Figure : 5.1

Consider the circuit diagram shown in figure 5.1 and assume the corresponding signals to be as following.

$m_1(t) = 10\cos 600t$, $m_2(t) = 10\cos 900t$, $c_1(t) = 20\cos 30000t$, $c_2(t) = 20\cos 60000t$ Now Answer the following questions.

- (a) Assume that, both the switches are on (circuit connected). Draw the frequency spectrum of $m_1(t)$ and $m_2(t)$. Determine the mathematical equations of $M_1(t)$, $M_2(t)$, $M(t)$ and $m(t)$. Draw the frequency spectrum of each of them too. (20)
- (b) Considering both switches being on, explain how it is possible to generate $m_1(t)$ as the output of the demodulation side. [You can use the derived equation of $m(t)$ from 5(a)] (5)
- (c) Assume that, the switch, S_1 is open and the carrier, $C_2(t)$ is also passed with $M_2(t)$. Draw the envelope of $M(t)$. Can this scheme be considered as a tone modulation now? Find out the modulation index, power efficiency under this constraint. (10)

6. (a) What is power spectral density? Briefly explain how a signal power is determined given its power spectral density. (5)

(b) What is Quadrature Amplitude Modulation (QAM)? Explain how the modulation and the demodulation schemes of a QAM signal work, including their circuit diagrams. Describe how the above-mentioned modulation-demodulation scheme can be extended to create a Quadrature Phase Shift Keying (QPSK) signal. (13)

(c) Draw the constellation diagram of the generic QPSK signal created by the scheme in 6(b). Consider the data stream: "10010110". Assume this data is passed using the QPSK signal above. Draw the bipolar NRZ form graph representing the Q-channel of the data stream. Mention the corresponding phase shift in the QPSK signal for each pair of clock cycles. Determine two timestamps with the lowest and the highest phase shift during transmission. (9)

(d) How does Offset-QPSK (OQPSK) benefit the phase shifting scenario of the QPSK scheme? For the same data stream of 6(c), how will you create a 16-ary PSK signal using the previous QPSK signal? What will be the minimum phase shift then? (8)

7. (a) A modulating signal is defined using the function: $m(t) = 3t - 6[0.5t]$ Volts, where $[t]$ is the integer part of t . Draw the modulating signal and determine its frequency in Hertz. The carrier signal is defined to be $c(t) = \cos(2\pi f_c t)$, $f_c = 10\text{kHz}$ Volts. Write the equation of a Phase Modulated (PM) signal, (where $c(t)$ is modulated by $m(t)$). Draw the graphs of the carrier signal and the PM signal Briefly explain the nature of phase shifting in your graph with respect to $m(t)$. (16)

(b) The equation of an FM signal is $m'(t) = 20\cos(5 \times 10^7 t + 4\sin(7 \times 10^3 t))$ volts. Determine the carrier and modulating frequencies, modulation index, and frequency deviation.

If the frequency sensitivity is 2 Hz/Volt, determine the modulating signal. (13)

Assume,

$$j_0 = -0.025 \times m_f, j_1 = -0.0175 \times m_f, j_2 = 0.09 \times m_f, j_3 = 0.1075 \times m_f, j_4 = -0.07 \times m_f, j_5 = 0.0325 \times m_f, j_6 = 0.0125 \times m_f, j_7 = 0.0325 \times m_f, j_8 = 0.005 \times m_f, j_n = 0.0, n > 7$$

Calculate the total power of the transmitted signal.

- (c) Discuss three differences between Amplitude Modulation and Angular Modulation. (6)

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8. (a) Show the significance of Hilbert transformation in achieving a Single Sideband Modulation. Explain using figures of frequency spectrum and mathematical analysis. (13)
- (b) You are assigned to arrange a pattern of lighting in a family get-together using 2 LEDs. The pattern should be as following: (8)
- During a period of 800 ms, the 1st LED will be lit for the first 200 ms, the 2nd one will be lit for the next 400 ms, and the 1st one will be lit again for the last 200ms.
- All you are provided with are: a sinusoidal signal, $s(t) = \cos 2\pi \times 1.25t$ Volts and two Op-Amps. Explain how you are going to avail the desired functionality using circuit diagrams for the 2 LEDs. Assume, to keep the LED lit, you have a 5 Volts dc supply.
- (c) What is Frequency Division Multiplexing (FDM)? Explain the benefit of FDM in handling various range of frequencies for a large range of communication channels. (7)
- (d) Show that, the minimum frequency difference to make two signals orthogonal is, (7)
- $$\delta f = \frac{1}{2T_b}$$
-

SECTION – AThere are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Define DC component problem in a line coding scheme. How does it cause problem while transmitting a signal through a communication channel? With appropriate examples show that both NRZ-L and NRZ-I exhibit the DC component problem. Why don't AMI and pseudo-ternary line coding schemes have this problem despite the existence of long sequence of 1's or 0's in a data stream? (25)
- (b) Can we foretell the future events of a signal from its past samples? Justify. (10)
2. (a) With necessary mathematical analysis, show that the receiver in differential pulse code modulation (DPCM) is able to produce the original message signal pulse the quantization noise. How does the signal-to-noise ratio (SNR) improve here though there is quantization noise? (18)
- (b) Calculate the coefficients of the exponential Fourier Series (FS) expansion of the signals $\sin 2\pi t$ and $\cos 2\pi t$. Comment on the characteristics of these signals and their corresponding coefficients in FS expansion. Do we need to integrate the signals over their complete cycles while calculating their coefficients in FS expansion? Justify. (17)
3. (a) A practical signal cannot be fully recovered from its sampled values no matter how much we increase the sampling frequency. Justify the statement. Which phenomenon causes this problem and how does it distort the signal? What do we do to minimize this distortion? (24)
- (b) State the Dirichlet condition for the existence of Fourier Transform (FT) of a signal. Is it a necessary condition of the existence of the FT of a signal? Justify with an appropriate example. (11)
4. (a) State the characteristics of distortionless transmission. With appropriate examples give physical interpretation of these characteristics. What happens if these characteristics are not satisfied? (20)

CSE 311**Contd ... Q. No. 4**

(b) In a hypothetical telemetry system, there are five analog signals. The bandwidth of the first one is 5.0 kHz, but for each of the remaining signals it is 2.5 kHz. These signals are to be sampled at rates no less than their respective Nyquist rates and no more than the Nyquist rate of the first one and are to be word-by-word multiplexed. This can be achieved by multiplexing the PAM samples of the five signals and then binary coding the multiplexed samples (as in the case of the PCM T1 carrier mentioned in your text book). **Suggest and draw** a suitable multiplexing scheme for this purpose. Note: In this case, you may have to sample some signal(s) at rates higher than their Nyquist rate(s). **What is the commutator frequency** (in rotations per second)? If the samples are quantized into 1024 levels, **determine the binary pulse rate** (bits per second) of the composite binary-coded signal, and **the minimum bandwidth** (in kHz) required to transmit this signal. (15)

SECTION - B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Max wants to design a 4-ary FSK scheme. He wants B/W consumption to be as small as possible while keeping the FSK signals orthogonal to each other in T_b . Derive the most appropriate frequency increment, δf , among signals which would fulfill Max's requirements. Finally, specify a set of FSK signals suited to the task when $T_b = 10$ ms. (11+4=15)
- (b) Consider the scheme designed in 5(a). Explain with necessary block diagram how the transmitted FSK signals can be detected at the receiver's end using Noncoherent and Coherent techniques. (5+5=10)
- (c) Sketch the PM wave corresponding to the modulating signal shown in Figure 5.1, given $f_c = 10$ MHz and $k_p = 5\pi$. (10)

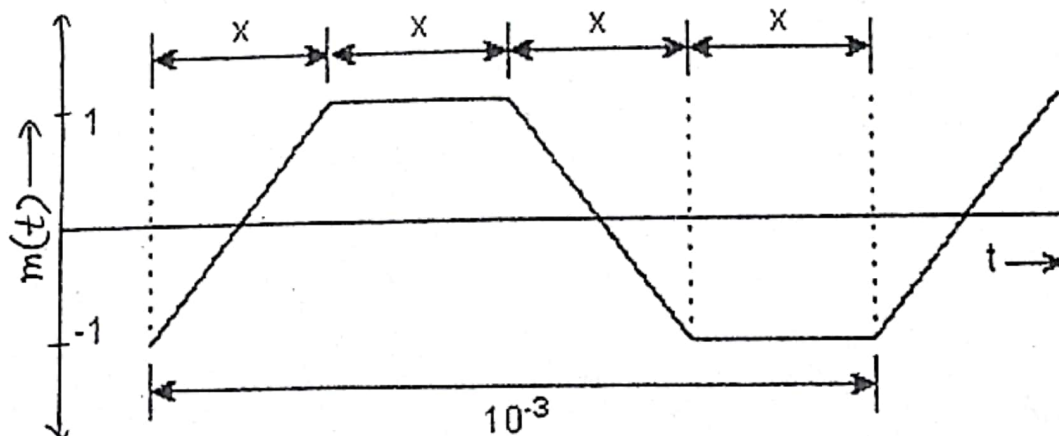


Figure 5.1

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6. (a) Rank the following digital modulation schemes in terms of probability of error. Provide necessary figures/diagrams, and explanations to support your claim. (3×6=18)
- (i) Binary ASK
 - (ii) Binary PSK
 - (iii) Binary FSK
- (b) Briefly explain when and why the value of $k_p m(t)$ has to be restricted within $(-\pi, \pi)$ during Phase Modulation. (5)
- (c) Suppose you want to design a DSB-SC modulator, where you simply multiply the modulating signal $m(t)$ with a carrier. But unfortunately, you do not have any multiplier at your disposal. Propose an alternate solution to achieve modulation. Show necessary diagrams. (12)
7. (a) Compute the estimated bandwidth of Wide Band FM signal using staircase approximation. Provide necessary figures. (15)
- (b) Explain why envelope detection does not work for PSK. Propose an alternate approach/modification to PSK so that envelope detection can be used during demodulation. (4+8=12)
- (c) Draw a block diagram for SSB modulation (Lower sideband only). (8)
8. (a) Suppose you want to transmit 4 bits of data at a time using M-ary PSK. Specify a set of PSK signals suited to the task. (4+8=12)
- Also compute the distance between two consecutive points in the signal space diagram, showing detailed steps.
- (b) Draw the block diagram for an NBFM modulator using a DSB-SC modulator. (10)
- (c) Explain how Offset-QPSK deals with the high phase shift problem of QPSK. (7)
- (d) Describe how a Frequency Modulated signal can be demodulated using Envelope Detection. (6)
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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

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Sub: **CSE 311** (Data Communication)

Full Marks: 210 Time: 2 hours and 30 minutes

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks.

SECTION – AThere are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Justify why the following signal can be represented using Fourier series. (17)

For students with ODD roll numbers: the signal $\cos 2\pi f_0 t$ **For students with EVEN roll numbers:** the signal $\sin 2\pi f_0 t$

Show whether the coefficients of the exponential Fourier series of your assigned signal are real or imaginary.

- (b) Represent your assigned signal in Question 1(a) using Fourier transform. Does this signal satisfy Dirichlet condition? Comment on the necessity of the Dirichlet condition for the existence of Fourier transform. (18)

2. (a) Show that time delay in a signal causes a **linear phase shift** in its spectrum. Explain what would happen in a voice signal consisting of multiple signals with different frequencies if this **linear phase shift** property does not hold. (24)

- (b) Prove that a unit step signal can be derived from a unit impulse signal and vice versa. (11)

3. **Find and draw** the Fourier transform of the following signal. (35)

For students with ODD roll numbers: the signal $2B\text{sinc}(2\pi Bt)$ **For students with EVEN roll numbers:** the signal $B\text{sinc}^2(\pi Bt)$ If your assigned signal is sampled at the rate of **1.5×Nyquist sampling rate**, **draw and mathematically represent** the spectrum of the sampled signal. Now derive the generic mathematical equation of the spectrum of a signal $g(t)$ after sampling it at the rate of f_s .

4. (a) Show that the signal-to-noise ratio (SNR) of a quantized signal is proportional to the square of the number of quantization levels (L). How is SNR improved in differential pulse code modulation? How does delta modulation ensure sufficient SNR despite that its number of quantization levels is only 2? (27)

- (b) Draw the graphs of the NRZ-I and 2B1Q schemes for the following data stream. Assume the last signal level has been positive. (8)

For students with ODD roll numbers: 0110 0110 0110 0110**For students with EVEN roll numbers:** 1100 1100 1100 1100

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Suppose, your roll number is x (the range of x is 1-121) and (35)

$$m(t) = \cos(2\pi x t) \cos(5\pi x t).$$

The carrier is $\cos(1000\pi t)$. Now, answer the following questions.

- i. Write down the exact equation for $m(t)$ in your particular case.
- ii. Draw the spectrum of $m(t)$ w.r.t. f .
- iii. Draw the modulation spectrum of DSB-SC w.r.t. f .
- iv. Draw the demodulation spectrum of DSB-SC w.r.t. f .
- v. Draw the modulation spectrum of SSB-SC (for LSB) w.r.t. f .

Note that, you need to show the necessary calculation steps before drawing the mentioned spectra.

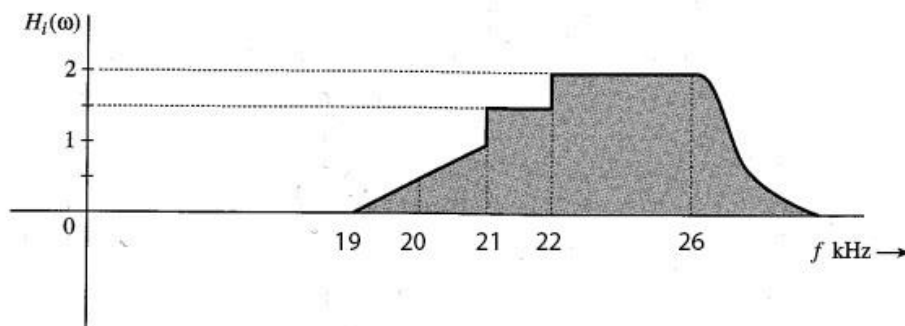
6. (a) Derive the equation for the minimum frequency gap in case of M-ary FSK. (15)

- (b) How does MSK work? Explain using an example. (10)

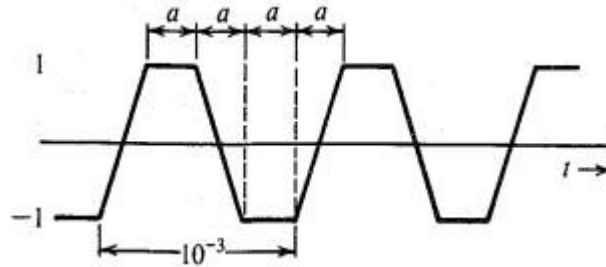
- (c) How are QPSK and Offset-QPSK different from MSK? (10)

7. (a) The carrier frequency of a certain VSB signal f_c is 20 kHz and the baseband signal (20)

bandwidth is 6 kHz. The VSB shaping filter $H_i(\omega)$ is shown below. Now, find the required output filter $H_o(\omega)$ for distortionless reception.



- (b) Derive the condition for envelope detection of SSB-WC. (10)
- (c) Draw the block diagram for a Phase Lock Loop. (5)
8. (a) Sketch PM and FM waves for the modulating signal $m(t)$ shown in the figure below. (20)
- Here, $\omega_c = 10^9$, $k_f = 10^5$, and $k_p = 25$.



- (b) What are the problems that might occur during the demodulation phase of QAM? (15)
- Explain it with the help of an appropriate example.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) A signal $g(t)$ band-limited to 50 Hz is uniformly sampled at intervals of 0.01 sec apart such that the following sampled values are found: (21)

$$g(0 \text{ sec}) = 1, g(\pm 0.01 \text{ sec}) = g(\pm 0.02 \text{ sec}) = g(\pm 0.03 \text{ sec}) = \dots = 0$$

Find and draw the signal $g(t)$ and its Fourier transform (FT). Write down the characteristics of $g(t)$. Can you justify the FT of a Dirac delta function from $g(t)$ and its FT?

(b) Characterize a distortionless transmission system. Justify whether an All-Pass transmission system is distortionless. (14)
2. (a) Justify the relationship between the sampling frequency and the quality of a delta modulator. Why and how does a delta modulator replace the delay component with an integrator? How does the threshold of coding have impacts on slope overload and granular noise in delta modulation? (20)

(b) Show that the coefficients of the exponential Fourier series of an even symmetric and an odd symmetric periodic signal are real and imaginary, respectively. Also show that the coefficients can be calculated by integrating the signal over the half cycle only. (15)
3. (a) The input stream to a 4B/5B block coder is 1100 0000 0000 0000 0011. What is the output stream? What are the lengths of the longest consecutive sequence of 0's in the input and the output streams? What will be the results of scrambling the input stream using B8ZS and HDB3 techniques? Assume that the last non-zero signal level has been positive and the number of non-zero pulses is odd after the last substitution. (Please see **Table for Q. No. 3(a)**). (15)

(b) Explain the main cause of aliasing which occurs during sampling a signal? What problems does aliasing introduce to a sampled signal? How are they compensated? (20)
4. (a) Establish the equation which shifts the spectrum of a signal $g(t)$ to a different frequency band. Explain how it is used to transmit multiple signals simultaneously over a shared channel. Also briefly introduce a dual technique that does the same job. (18)

(b) Prove that for an input signal $x(t)$, the output of a linear time invariant system with impulse response $h(t)$ is given by $y(t) = x(t) * h(t)$. Use ONLY time domain variables and equations. (17)

Data Sequence	Encoded Sequence	Control Sequence	Encoded Sequence
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Table for Q. No. 3(a)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) An Amplitude Modulated (AM) wave is represented by the expression: (10)

$$v = 5[1 + 0.6 \cos(6280t)] \cdot \sin(2\pi \times 10^4 t)$$

Calculate (i) Modulation depth, (ii) modulating frequency, (iii) period of carrier wave, (iv) peak instantaneous value of the modulated wave.

(b) Consider that one input to an AM DSBFC modulator is an 800 KHz carrier with an amplitude of 40 V. The second input is a 25 KHz modulating signal whose amplitude is sufficient to produce a $\pm 10V$ change in the amplitude of the envelope. Calculate (i) upper and lower side band frequencies, (ii) Modulation coefficient and percentage modulation, (iii) Maximum and minimum positive amplitude of the envelope. In addition, draw the envelope and the output frequency spectrum. (15)

(c) Show that in amplitude modulation, maximum average power transmitted by an antenna is 1.5 times the carrier power. Also, show that for 100% modulation, each sideband contains one-sixth of the total power. (10)

6. (a) Why is it possible to exceed 100 percent modulation in frequency modulation, but not in amplitude modulation? Derive the mathematical expression for single tone frequency modulation. (12)

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Contd..... Q. No. 6

- (b) What is the bandwidth required for an FM signal if the modulating frequency is 1 KHz and the maximum deviation is 10 KHz. What is the bandwidth required for a DSBFC amplitude modulation? (8)
- (c) What is the Carson's rule? Use Carson's rule to compare the bandwidth that would be required to transmit a baseband signal with a frequency range from 300 Hz to 3 KHz using (i) NBFM with maximum deviation of 5 KHz (ii) WBFM with maximum deviation of 75 KHz. (7)
- (d) In an FM system, the audio frequency is 1 KHz and audio voltage is 2V. The deviation is 4 KHz. If audio voltage is now increased to 8 volts and audio frequency is dropped to 500 Hz, find the modulation index in each case and the corresponding bandwidth. (8)
7. (a) Draw the constellation diagram of quadrature phase shift keying (QPSK). Write the mathematical expression of a QPSK signal. Explain the working operation of an offset QPSK (OQPSK) transmitter. (10)
- (b) Draw the geometric representation of M-ary PSK signals and find the distance between signal points. Also find the bandwidth of M-ary PSK. (10)
- (c) Given an 8-level PSK signal that employs two different amplitudes as shown in figure 7(c). Calculate the distance between two nearest signal points and compare with the minimum distance for standard 8-level PSK. Which scheme is better in terms of error probability? (10)

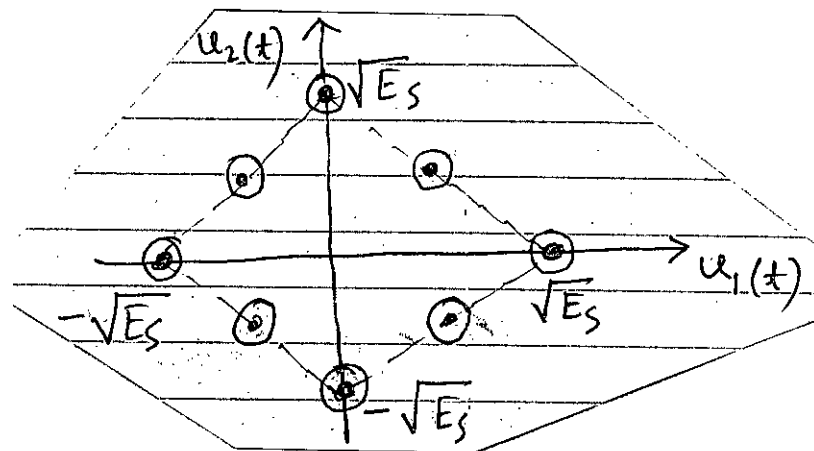


Figure for Q. 7(c)

- (d) Assume that you are required to transmit $f_b = 90$ Mb/S data in an authorized bandwidth of 20 MHz using M-ary PSK. Find the number of bits per symbol. (5)
8. (a) Write the mathematical expression of amplitude shift keying (ASK) waveform. Why is bandpass filter used in generation of ASK waveform? (6)
- (b) What are the desirable characteristics of a line coding scheme? Describe the Manchester and the differential Manchester line coding techniques. (10)
- (c) Consider a binary sequence: 11111011111. Draw the waveforms for (i) Unipolar NRZ, (ii) Bipolar NRZ, (iii) AMIRZ. Discuss the advantages and disadvantages of the three signaling formats. (10)
- (d) Explain why BPSK signaling scheme is better than BFSK signaling in terms of error probability? (9)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

All the symbols have their usual meanings unless explicitly mentioned.

1. (a) Why are we interested in frequency domain representations of signals? How do we obtain frequency spectrum of a signal from its time domain representation? Write the corresponding formula. What is the difference between frequency spectrum of a periodic signal and frequency spectrum of an aperiodic signal? Explain graphically using the examples of a rectangular pulse and a periodic rectangular pulse train. Does being periodic or aperiodic have any effect on the bandwidth of a signal? Explain using the same examples, i.e., a rectangular pulse and a pulse train of the same pulse duration, τ . (4+3+6+5)

- (b) Let $G(f)$ denotes the Fourier transform of the rectangular pulse $g(t)$ in Figure (a) below. Now, using properties of Fourier transform, express the Fourier transform of the triangular pulse $h(t)$ in Figure (b) in terms of $G(f)$. Show all steps of calculation. (10)

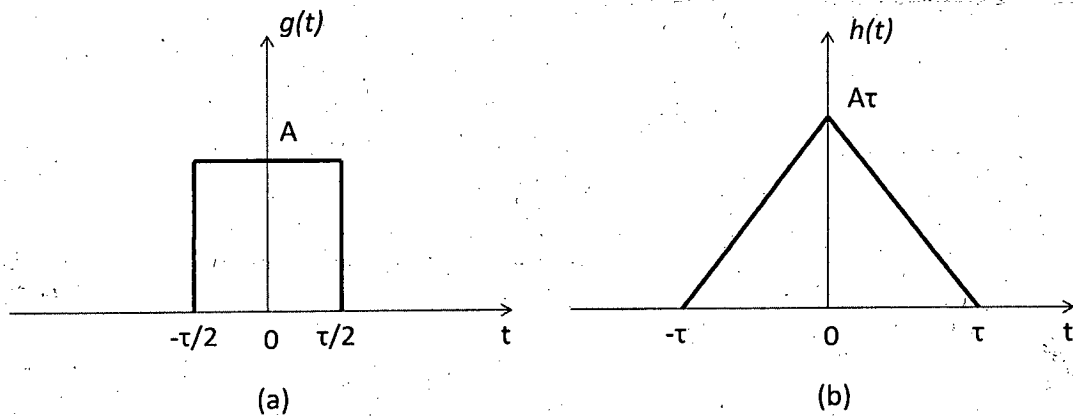


Figure for Question 1(b)

- (c) Write the transfer function of an ideal low-pass filter. From the transfer function, derive the impulse response, in time domain, of the ideal low-pass filter. (2+5)
2. (a) What is power-bandwidth trade-off? Demonstrate power-bandwidth trade-off with an example. (4+6)
- (b) Show the Signal-to-Noise Ratio (SNR) of a uniformly spaced quantizer is proportional to input signal power. Assume all values in the amplitude range of the quantizer are equally likely. (10)

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Contd... Q. No. 2

(c) A PCM system implements μ -law companding where $\mu = 255$. Total number of levels in the quantizer is $L = 256$. Now, for each of the following sample values: **-2.4, 0.6 and 5**, what will be the value received after expansion. The μ -law (for positive amplitude) is given by:

(15)

$$y = \frac{1}{\ln(1+\mu)} \ln \left(1 + \frac{\mu m}{m_p} \right), \quad 0 \leq \frac{m}{m_p} \leq 1$$

Assume $m_p = 5$. Show detailed calculation steps.

3. (a) Draw a block diagram of the transmitter of a DPCM (Differential Pulse Code Modulation) system. Delta Modulation (DM) system is a special case of DPCM. What are the special features of a DM system with respect to a general DPCM system? What is the advantage of using a DM system?

(5+3+3)

(b) Given a sequence of samples $\{g_0, g_1, \dots, g_{N-1}\}$ of an analog signal $g(t)$, the Discrete Fourier transform (DFT) of the sequence is defined as:

(10+8+6)

$$G_k = \sum_{n=0}^{N-1} g_n e^{-j \frac{2\pi}{N} kn}, \quad k = 0, 1, \dots, N-1$$

Here the sequence $\{G_0, G_1, \dots, G_{N-1}\}$ is called the transform sequence.

Now formulate a *decimation-in-frequency* FFT (Fast Fourier Transform) algorithm to efficiently compute the DFT of a finite data sequence. Draw the complete *signal-flow* graph corresponding to the computation of an 8-point DFT (i.e., $N = 8$) using your algorithm. If *in-place computation* is used for space efficiency in such an algorithm, then the samples of the transform sequence G_k are stored in a *bit-reversed order*. Explain using the *signal-flow* graph drawn, what *bit-reversed order* means.

4. (a) Describe two causes of asynchronicity among incoming channels of a TDM (Time Division Multiplexing) multiplexer. Describe how this problem of asynchronicity is resolved in a TDM system.

(6+9)

(b) What is 'Digital Line Coding' technique? Describe six desirable characteristics of a line coding technique. For each of the characteristics, give examples: names and encoding rules of two line coding schemes such that one is better than the other with respect to that characteristic. Justify your examples with proper explanation.

(2+18)

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SECTION – B

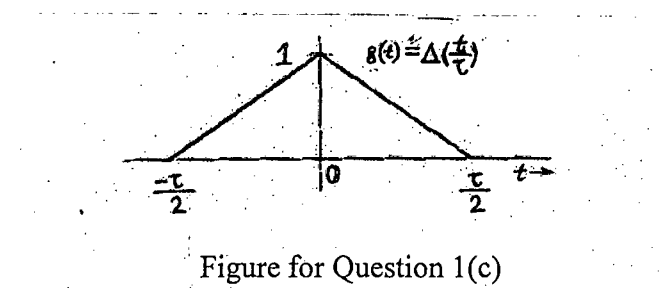
There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Mention the transmission bandwidths required for DSB, QAM and SSB modulation techniques. (3+5+5+5+5=23)
- For a base signal $m(t) = \cos 100t \times \cos 200t$
- (i) Draw the spectrum of $m(t)$.
 - (ii) Draw the spectrum of DSB-SC for $2m(t)\cos 100t$.
 - (iii) Draw the spectrum of DSB+C for $2(1+m(t))\cos 1000t$.
 - (iv) Draw the spectrum of LSB-SC from the figure you draw for Question (ii)
- (b) How does the value of RC affect Envelope Detection in DSB-WC? Mention the condition imposed on the value of RC. (8+4=12)
6. (a) Define Tone. What do you mean by Tone Modulation? (3)
- (b) Draw a detailed block diagram for SSB Modulation (Upper Side band only). (15)
- (c) Write the modulation equations for Frequency Modulation and Phase Modulation. Which one is better in practice? Draw the block diagrams for converting a Frequency Modulator into a Phase Modulator and vice versa. (8+2+7=17)
7. (a) Define PSK and DPSK. What is the difference between the two? (6+2=8)
- (b) For Vestigial Sideband Modulation, derive the following equation for an equalizer filter: $H_o(f) = \frac{1}{H_i(f+f_c) + H_i(f-f_c)}$ (for an input filter $H_i(f)$, and carrier frequency f_c). Draw the figures necessary for derivation. (27)
8. (a) Draw a detailed block diagram for a Nonlinear DSB-SC Modulator. (15)
- (b) Mention the bandwidth requirements for NBFM and WBFM. Draw the block diagram for constructing an NBFM Modulator using a DSB-SC Modulator. (5+15=20)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) Define bandwidth of a signal. Sketch two time-domain signals such that bandwidth of one signal is greater than the other. Justify your answer using the definition of bandwidth (4+5=9)
- (b) Prove that for a real-time signal, the amplitude spectrum is an even function and the phase spectrum is an odd function. (6)
- (c) Let $\Delta(\frac{t}{\tau})$ represents a triangular pulse as shown in the figure below. The Fourier transform of $\Delta(\frac{t}{\tau})$ is given as: $\Delta(\frac{t}{\tau}) \Leftrightarrow \frac{\tau}{2} \text{sinc}^2(\frac{2\pi f\tau}{4})$. Now, using duality property, find the Fourier transform of the time function: $\text{sinc}^2(t)$ in terms of the function $\Delta(\cdot)$. Also, draw the spectrum. (6)



- (d) Give a proof for why an ideal low pass filter is physically unrealizable. (8)
- (e) Consider a Linear Time Invariant (LTI) system that functions as a differentiator, i.e., given a signal $g(t)$ as input, its output will be $\frac{d}{dt} g(t)$. Derive and write down the transfer function of the system. (6)
2. (a) List three important advantages digital communication over analog communication. (5)
- (b) State and prove Nyquist's sampling theorem. In light of the proof, explain why in practice – (i) a signal is passed through a band-Limiting filter before sampling, and (ii) sampling rate is higher than the Nyquist rate. (10+6=16)

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Contd ... Q. No. 2

(c) Describe how companding achieves non-uniform quantization without actually changing the uniform level spacing in the quantizer. Explain intuitively how a quantizer with companding causes Signal-to-Noise Ratio (SNR) to be independent of input power. (9+5=14)

3. (a) Let $G(f)$ be the Fourier transform of a real-time signal $g(t)$. In discrete Fourier Transform (DFT), we compute $G(f)$ at some finite frequencies from the samples of $g(t)$. Let $g(t)$ be sampled over the duration $0 \leq t \leq T_0$ at uniform intervals of T_s seconds resulting in N_0 samples. Then the formula for computing DFT is as follows: (4+5+5+6)

$$G_q = \sum_{k=0}^{N_0-1} g_k e^{-jq\Omega_0 k}, \text{ where } g_k = T_s g(kT_s), G_q = G(qf_0), \Omega_0 = 2\pi f_0 T_s$$

Now answer the following.

(i) Derive the above formula from the expression of $G(f)$.

(ii) Prove that, $f_0 = \frac{1}{T_0}$.

(iii) Compute DFT for the following sample values of $g(t)$. Write down the values of sample frequencies, f and computed values of $G(f)$.

t	0	1	2	3
g(t)	1	2	3	4

(iv) What considerations should be made while selecting values for T_s and T_0 ? Explain.

(b) How does a good 'predictor' design improve the performance of a Differential Pulse Code Modulation (DPCM) system? Explain. (6)

(c) Consider a signal, $m(t) = A \cos \omega t$ is input to a Delta Modulation (DM) system. What should be the step size, E such that no slope overload occurs in the system for $m(t)$? How the step size can be made adaptive in a DM system? (5+4)

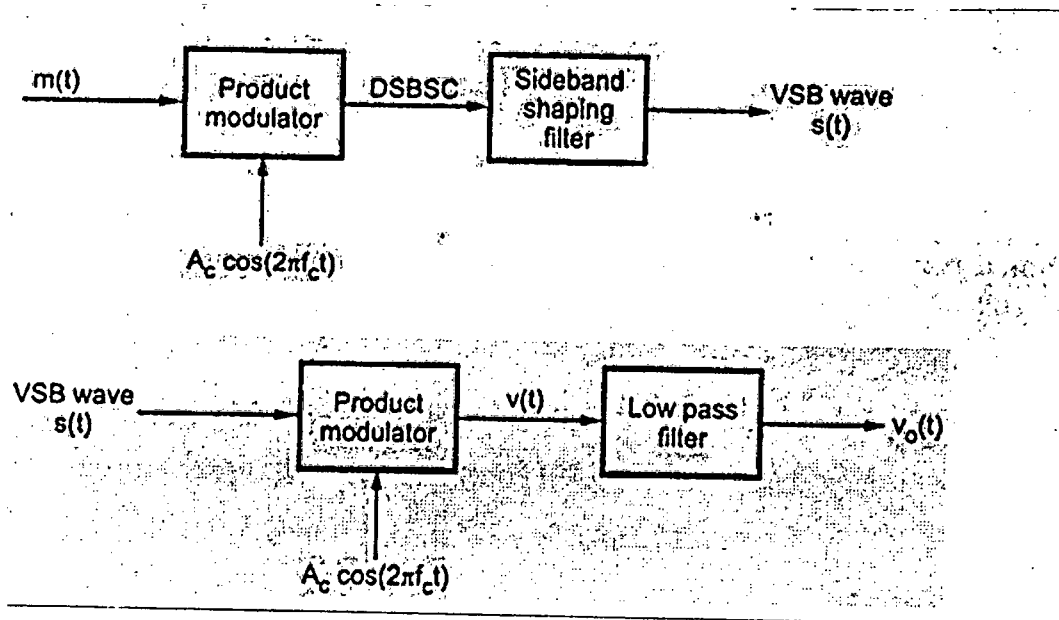
4. (a) Give a real life example where Time Division Multiplexing (TDM) is used. Why pulse/bit stuffing may be required in a TDM multiplexer? Explain the difference between positive pulse stuffing and negative pulse stuffing? How does the receiver detect a stuffed bit in each case of positive and negative pulse stuffing? (2+5+6+4)
- (b) Why does 'Inter-symbol Interference (ISI)' occur in a digital transmission system? Give an example of a pulse of bandwidth B Hz that can be used to transmit $2B$ pulses per second in presence of ISI. Explain how it works. How such a pulse can be generated using a transversal filter? Explain using a block diagram and necessary illustrations. (4+8+6)

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Suppose, an AM signal is described by $[A+m(t)] \cos \omega_c t$, Where $A \gg m(t)$. Show that, the message $m(t)$ can be recovered at the receiver from this AM signal by squaring it and then passing the resulting signal through a low-pass filter. (15)
- (b) Explain the necessity of Costas phase-locked loop in the demodulation of DSB-SC signals with a suitable example. (10)
- (c) Explain the operation of Costas phase-locked loop with approximate block diagram. (10)
6. (a) Derive the time domain representation of SSB signal with appropriate figures. Then based on this representation, design a SSB modulation technique using block diagram. (15)
- (b) The modulator and demodulator for VSB signal are shown below: (20)



Now derive the required characteristics of the transfer function $H(f)$ of the sideband shaping filter of modulator with appropriate figures

7. (a) Derive the bandwidth estimation of WBFM signal with appropriate figures using staircase approximation approach. (15)
- (b) Consider a FM signal $\cos \left[\omega_c t + k_f \int_0^t m(\alpha) d\alpha \right]$.

Draw a block diagram showing necessary components to convert the signal into

$$\cos \left[\omega_c t + 3k_f \int_0^t m(\alpha) d\alpha \right]. \quad (20)$$

And also for each component, clearly mention its function along with corresponding input and output signals.

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8. (a) What is 'Digital Line Coding'? Describe using examples how digital line coding techniques determine the following characteristics of transmitted signals. **(15)**
- (i) Bandwidth consumption
 - (ii) Power consumption
 - (iii) Synchronization Capability
 - (iv) DC value suppression capability
- (b) Mathematically prove that a FM modulator can be constructed using a PM modulator. Draw the block diagram of such FM modulator. **(10)**
- (c) How can we improve the spectral efficiency of amplitude modulation using Quadrature Amplitude Modulation (QAM) technique? Explain with the block diagram of appropriate modulator and demodulator. **(10)**
-

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Give your answer to the point avoiding unnecessary writings.

Try to draw as many figures as possible and explain with block diagrams.

1. (a) Briefly explain the steps required for "Echo Cancellation"? Explain how "Inter-Symbol Interference (ISI)" can occur? (7+6)
- (b) Briefly explain the procedure of 4B/5B encoding scheme and intentions behind it. (10)
- (c) Show with necessary figures how 4-bit information is passed at a time with 16-QAM (Quadrature Amplitude Modulation). (6)
- (d) For a baseband signal of $m(t) = \cos 100t + \cos 400t$ (6)
 - (i) Draw basic spectrum of $m(t)$
 - (ii) Draw spectrum of the modulated signal using Double Side Band with Carrier (DSB+C), where the carrier signal is $2(1 + m(t))\cos 1000t$. (6)
2. (a) Draw digital signals of the corresponding binary sequence of 1011001 for the following line encoding schemes, (9+6)
 - (i) Pseudo-Ternary
 - (ii) Differential Manchester
 - (iii) Multi-transition (MLT)-3

Summarize each of these in terms of Bandwidth, Baseline wandering, and DC component issues.
- (b) Explain the incoherent demodulation procedure of FM and FSK. Explain how modulations of Phase Shift Keying (PSK) and Quadratic Amplitude Modulation (QAM) relate to each other? (5+3)
- (c) Explain how offset QPSK (Quadrature Phase Shift keying) reduces maximum phase shift from π to $\frac{\pi}{2}$ without reducing bit rate. Explain how phase continuation is maintained during MSK (Minimum shift keying). (6+6)

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3. (a) Show that, for a real time $f(t)$, the signal $f_{USB}(t) = f(t)\cos\omega_c t - f_h(t)\sin\omega_c t$ represents an upper sideband SSB-SC signal, where $f_h(t)$ represents Hilbert Transform of $f(t)$ and ω_c is the carrier frequency in rad/sec. (13)
- (b) What is orthogonal signal? Explain the motivations behind making each m-FSK signals orthogonal to each other? Show that the minimum frequency difference to make signal orthogonal is, $\delta_f = \frac{1}{2T_b}$ Hz. (2+4+6)
- (c) What was the fallacy behind the estimation of the bandwidth of Frequency Modulation (FM)? Show that the bandwidth requirement of Wide Band Frequency Modulation (WBFM) is $\approx 2(\Delta f + 2B)$ Hz, where Δf refers to peak frequency deviation. (4+6)
4. (a) Explain with diagram how Phase Locked Loop (PLL) maintains Phase and Frequency synchronization between source and generated carrier signals. (15)
- (b) For a Vestigial Side Band (VSB) modulation with suppressed carrier, derive following relation between input and output filter, $H_o = \frac{1}{H_i(\omega + \omega_c) + H_i(\omega - \omega_c)}$. (8)
- (c) Explain with necessary figures and diagrams, the modulation and demodulation techniques of On Off Keying (OOK), Frequency Shift Keying (FSK), and Differential Phase Shift Keying (DPSK). (4+4+4)

SECTION-B

There are **FOUR** questions in this section. Answer any **THREE**.

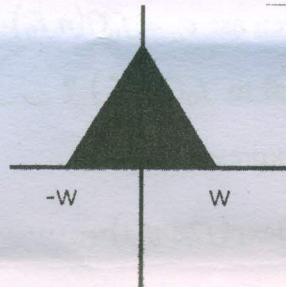
5. (a) There are two principal forms of digital switching in communication networks, (10)
- Circuit switching
 - Packet switching
- Big website companies like Facebook and Google store their data in data centers where thousands of computers act together as a single server. Also, they establish multiple data centers in geographically distance locations. That means when you load your Facebook home page, it may be the case that the picture of Aashik Salam, one of your computer science teachers, was downloaded from the Facebook data center at Singapore and the picture of Himel Sen, another of your computer science teachers, was downloaded from the data center at California. However, it is not just the case that only clients communicate with the data center servers, data centers communicate with each other a lot as well. What form of digital switching do these data centers use to communicate with each other?
- (b) State and prove the dilation property of the Fourier transform. (5+5)
- (c) If the Fourier Transform of the signal $g(t)$ is $G(f)$, then determine the Fourier Transform of the signal, $\sin^2(2\pi f_c t)g(t)$. (10)
- (d) Derive the Fourier Transform of the Dirac Delta Function. (5)

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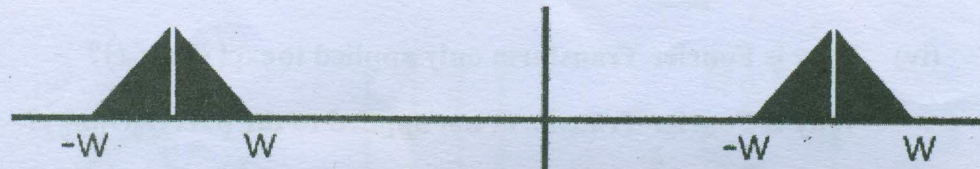
6. (a) Determine the bandwidth of the following two signals. The frequency domain representation of the signals are drawn as follows. Also tell what will be the bandwidth if the diagrams drawn are for time domain representation.

(3+3+4)

(i) Signal One



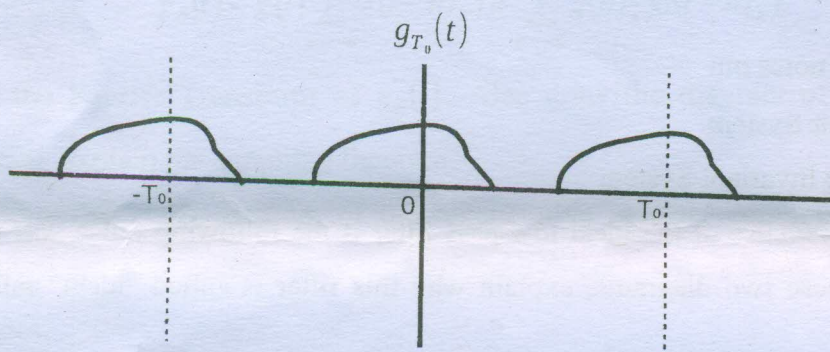
(ii) Signal Two



(b) Derivation of the Fourier Transform of periodic signals are given below. Read the derivation given below and write answer the questions:

(20)

Let $g_{T_0}(t)$ be a periodic signal with period T_0 . The diagram and the definition of $g_{T_0}(t)$ regarding complex exponential Fourier Series are given below,



$$g_{T_0}(t) = \sum_{n=-\infty}^{\infty} c_n \exp(j2\pi n f_0 t) \dots \dots \dots (1)$$

where c_n is defined as

$$c_n = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} g_{T_0}(t) \exp(-j2\pi n f_0 t) dt \dots \dots \dots (2)$$

Now, a pulse like function $g(t)$ is defined in terms of $g_{T_0}(t)$ as,

$$g(t) = \begin{cases} g_{T_0}(t) & ; -\frac{T_0}{2} \leq t \leq \frac{T_0}{2} \\ 0 & ; \text{elsewhere} \end{cases} \dots \dots \dots (3)$$

(i) Why do we need to introduce this new function $g(t)$? [Given: the Fourier Transform of $g(t)$ if $G(f)$]

(ii) Also, Draw the time domain diagram of $g(t)$.

Now c_n can be written as: $c_n = f_0 G(n f_0)$ (4)

(iii) How can c_n be equal to $f_0 G(n f_0)$?

Now $g_{T_0}(t)$ can be expressed as:

$$g_{T_0}(t) = \sum_{n=-\infty}^{\infty} f_0 G(n f_0) \exp(j 2 \pi n f_0 t) \dots \dots \dots (5)$$

and the Fourier Transform of $g_{T_0}(t)$ would be,

$$F[g_{T_0}(t)] = \sum_{n=-\infty}^{\infty} f_0 G(n f_0) F[\exp(j 2 \pi n f_0 t)] \dots \dots \dots (6)$$

(iv) Why is Fourier Transform only applied to $\exp(j 2 \pi n f_0 t)$?

Why is Fourier Transform not applied to the part $f_0 G(n f_0)$?

From this definition we can conclude,

$$F[g_{T_0}(t)] = f_0 \sum_{n=-\infty}^{\infty} G(n f_0) \delta(f - n f_0) \dots \dots \dots (7)$$

(v) How can we deduce Eq. 7 from Eq. 6?

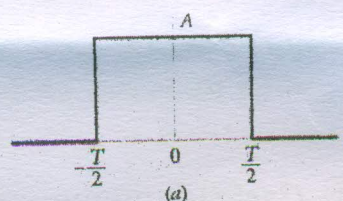
(c) Why is non-uniform quantization needed? What are the problems in case of uniform quantization? (5)

7. (a) Write short notes on- (5+5)

(i) Linear System

(ii) Time Invariant System

(b) Draw the response of the ideal low-pass filter if the following pulse was given as input. From these two diagrams, explain why this filter is called "ideal" rather than "real"? (10)

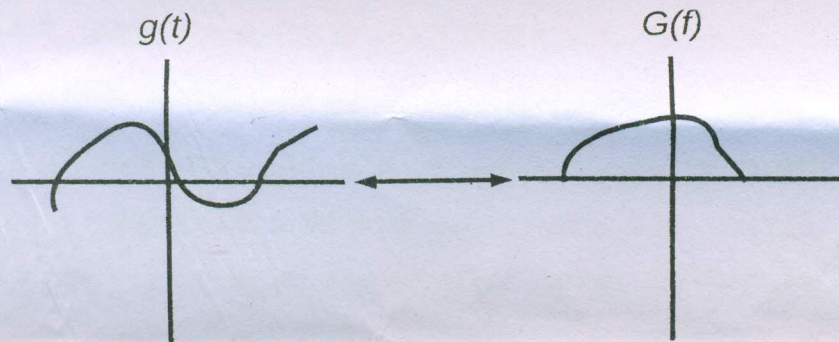


(c) Deduce the recurrence relationship for the Fast Fourier Transform. (15)

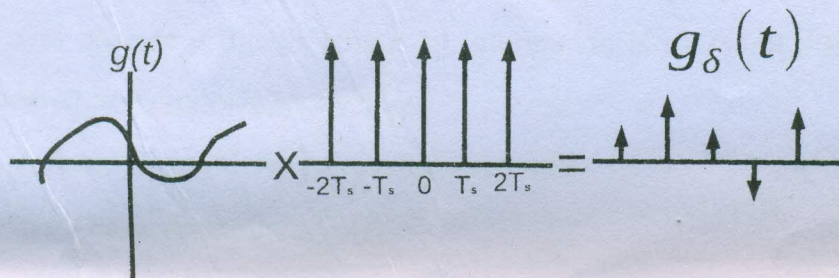
CSE 311

8. (a) Suppose a Fourier Transform pair signal is given as follows:

(6+3+6)



The signal is sampled as described in the following figure,



The sampled signal is described as,

$$g_{\delta}(t) = g(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s)$$

Find the Fourier Transform of $g_{\delta}(t)$. Also draw the diagram of $G_{\delta}(f)$. From the diagram, explain the sampling theorem.

(b) Draw the block diagram of the regenerative repeater component of Pulse code modulation. Also explain how each of the sub-components work in the regenerative repeater.

(10)

(c) What are the two types of error in case of delta modulation? Illustrate each of them with figures. Also, describe how these two type of errors can be overcome.

(10)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

All the symbols have their usual meanings unless explicitly mentioned.

1. (a) How is the bandwidth of a signal calculated? Accordingly calculate the bandwidth of the unit impulse function, $\delta(t)$ defined below. (5+7)

$$\delta(t) = 0 \quad t \neq 1$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

- (b) Explain 'Linearity' and 'Time invariance' properties of a 'Linear Time Invariant (LTI)' system. (10)

- (c) A low-pass filter transfer function $H(f)$ is given by (13)

$$H(f) = \begin{cases} (1 + k \cos 2\pi f T) e^{-j2\pi f t_d} & |f| < B \\ 0 & |f| > B \end{cases}$$

A pulse $g(t)$ band-limited to B Hz is applied at the input of the filter. Show that the output

$$y(t) = g(t - t_d) + \frac{k}{2} [g(t - t_d - T) + g(t - t_d + T)]$$

2. (a) List three advantages of digital communication over analog communication. (5)
- (b) A signal band limited to 4000 Hz is transmitted by a PCM system. The signal is sampled at twice the Nyquist rate and the sampled pulses are passed through a uniformly spaced quantizer with sixteen levels. Finally, the quantized samples are encoded as binary signals. Calculate the output data rate (in bits/sec) of the PCM system. (8)
- (c) Explain the 'Aliasing effect' that occurs during sampling of practical signals. How can 'Aliasing effect' be eliminated? (8+2)
- (d) Consider a PCM system that implements μ -law companding with $\mu = 255$. Total number of levels in the quantizer is $L = 256$. Now, what will be the total root-mean-square (rms) quantization error for the following set of sample values of some signal: $\{-2.4, 0.6, 5\}$. Assume the maximum and the minimum values of the signal are 5 and -5 , respectively. Show detailed calculation steps. (12)

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3. (a) Draw the block diagrams of transmitter and receiver of a Differential PCM (DPCM) system. Draw the block diagram of a linear predictor for DPCM. (8+5)
- (b) Draw the block diagrams of the modulator and the demodulator of a Delta Modulation (DM) system that use integrator-amplifiers and comparators. (8)
- (c) What is 'slope overload noise' in a Delta Modulation (DM) system? How to determine the step size of a DM modulator such that 'slope overload' does not occur for a given input signal $m(t)$? (5+9)
4. (a) What is Time Division Multiplexing (TDM)? Describe briefly how the following two issues are addressed in a system using TDM: (i) Synchronization between the sender (multiplexer) and the receiver (demultiplexer), (ii) asynchronicity among incoming channels in the multiplexer. (3+6+6)
- (b) Why does 'Inter-Symbol Interference(ISI)' occur in a digital transmission system? Explain how Nyquist first criterion for pulse shaping can be used to eliminate ISI. Does the pulse $p(t) = \text{sinc}(\pi R_b t)$ satisfy the criterion? If so, what is the maximum pulse rate achievable with $p(t)$? Explain. Accordingly, how much bandwidth (in Hz) would be required to transmit 4000 pulses per second using $p(t)$. Given, bandwidth of $p(t)$ is $\frac{R_b}{2}$. (5+6+4+3+2)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

Give your answer to the point avoiding unnecessary writings.

Try to draw as many figures as possible and explain with block diagrams.

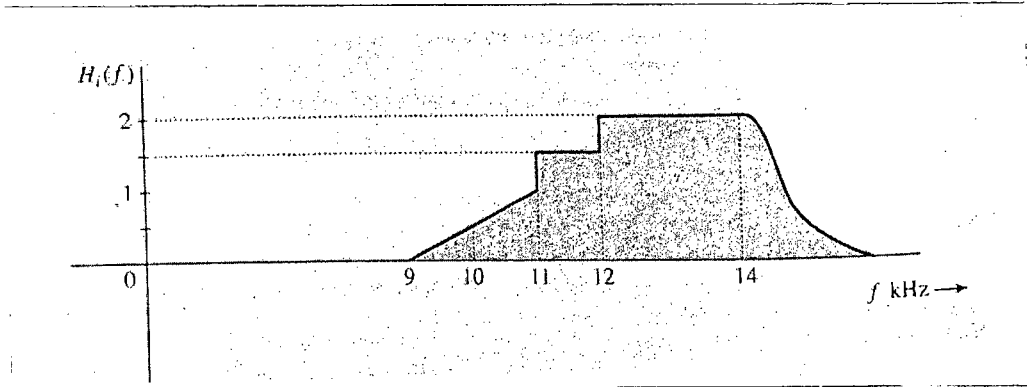
5. (a) For a base signal of $m(t) = \cos 100t \times \cos 200t$ (3+3+3+3+3=15)
 - (i) Draw spectrum of $m(t)$
 - (ii) Draw spectrum of DSB-SC for $2m(t)\cos 100t$
 - (iii) Draw spectrum of DSB+C for $2(1 + m(t))\cos 100t$
 - (iv) Draw spectrum of LSB-SC from the figure of (ii).

Just mention the transmission bandwidth required for DSB-SC, DSB+C and VSB+C modulation techniques.
- (b) Show that the bandwidth requirement of WBFM is $\approx 2(\Delta f + 2B)$ Hz, where Δf refers to peak frequency deviation. What was the fallacy behind bandwidth estimation of FM? (8+5)
- (d) Explain the modulation and demodulation procedure of Differential Phase Shift Keying (DPSK)? (7)

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6. (a) A vestigial filter $H_i(f)$ has a transfer function shown below. The carrier frequency is $f_c = 10$ kHz and the baseband signal bandwidth is 4 kHz. Draw the corresponding transfer function of the equalizer filter, $H_o(f)$. [Hint: $H_o(f) = \frac{1}{H_i(f + f_c) + H_i(f - f_c)}$.]

(10)



- (b) Show that for M-ary FSK, the minimum frequency separation to ensure orthogonality among the FSK symbols is: $\delta f = \frac{1}{2T_b}$, where T_b denotes the symbol duration.

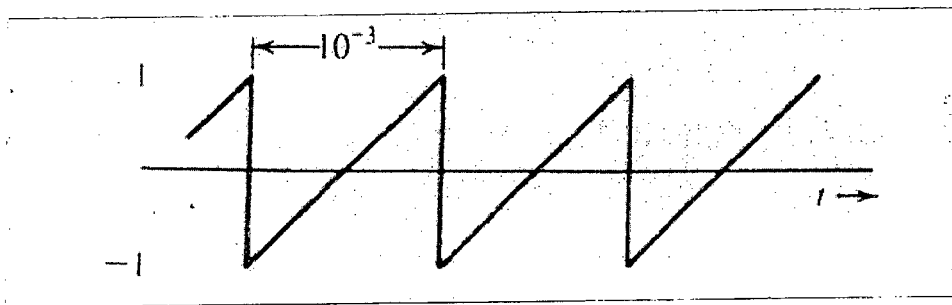
(11)

- (c) Show that, for a real time function $f(t)$, the signal, $f_{LSB}(t) = f(t)\cos\omega_c t + f_h(t)\sin\omega_c t$ represents a lower sideband SSB-SC signal $f_h(t)$ represents Hilbert Transform of $f(t)$ and ω_c is the carrier frequency in rad/sec.

(14)

7. (a) A baseband signal $m(t)$ is shown below. Sketch the $\phi_{FM}(t)$ and $\phi_{PM}(t)$ for this signal $m(t)$ if $\omega_c = 2\pi \times 10^6$, $k_f = 2000\pi$ and $k_p = \frac{\pi}{2}$. (Mention only the maximum and minimum frequency. No need to put extra effort on drawing.)

(10)



- (b) What is Pilot Carrier? Explain with diagram how Phase Locked Loop (PLL) maintains Phase and Frequency synchronization?

(2+10)

- (c) Explain with diagram how 4-bit information is passed at a time with 16-QAM?

(10)

- (d) Describe a trivial demodulation procedure of FM signal?

(3)

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8. (a) Draw the signals obtained after encoding the bit stream '10010110' by each of the following digital line coding techniques,

(6+9)

- Polar NRZ-L
- Manchester

Analyze these techniques in terms of the following characteristics.

- (i) Synchronization capability
- (ii) Bandwidth requirement
- (iii) DC value suppression

- (b) Explain how offset QPSK reduces maximum phase shift from π to $\frac{\pi}{2}$ without reducing bit rate? Explain how phase continuation is maintained during MSK, what are the prerequisites for carrier frequency to work with MSK.

(6+8)

- (c) What is baseline wandering? Describe how 'Scrambling' solves the synchronization problem with Bipolar-AMI.

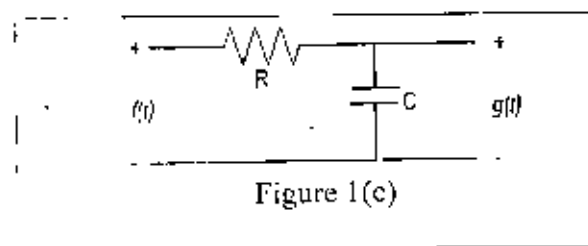
(2+4)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

All the symbols have their usual meanings unless explicitly mentioned.

1. (a) What are the advantages of using digital signals over analog signals? Explain. (5)
- (b) (i) How is the bandwidth of a signal calculated? (4+8)
- (ii) The time-scaling property of Fourier Integral is as follows:
If $g(t) \Leftrightarrow G(f)$, then for any real constant a , $g(at) \Leftrightarrow 1/|a| G(f/a)$.
Now which of $g(t)$ and $g(at)$ has higher bandwidth when $a > 1$? Explain your answer
- (c) (i) For a Linear Time Invariant (LTI) continuous time system, an input $x(t) \Leftrightarrow X(f)$ and the corresponding output, $y(t) \Leftrightarrow Y(f)$ are related as follows: $Y(f) = H(f)X(f)$ where $H(f)$ is called the transfer function of the system. Now given an LTI system, how would you empirically find out the transfer function of the system? Explain your answer. (8)
- (ii) For a 'Distortionless' LTI system, the amplitude response $|H(f)|$ must be constant and the phase response $\angle H(f)$ must be a linear function of f going through the origin, $f = 0$.
Now, under what conditions, the RC circuit in Figure 1(c) can be approximated as a Distortionless system. (10)



2. (a) State and prove Nyquist's sampling theorem. (3+10)
- (b) What are the practical issues in signal sampling and reconstruction according to the proof you have described above? How are those issues handled in practice? (3+7)
- (c) Explain how quantization with non-uniform spacing of levels can be achieved without actually changing the uniform level spacing in the quantizer of a PCM system? (12)

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3. (a) What are the special features of a Delta Modulation (DM) system in terms of (i) sampling rate, (ii) number of levels in quantizer, and (iii) predictor implementation? (3)
- (b) Draw the block diagrams of the modulator and the demodulator of a Delta Modulation system. (4+4)
- (c) The following set of sampled values of transmitted through the modulator of the DM system you have just drawn above. Compute the values received (at the demodulator) for each of the samples. Also compute the total RMS noise. Assume, step value, $E = 0.5$. Make any other assumptions required (6+4)
- Sampled Values: 0.2, 0.3, 0.4, 0.4, 0.8, 1.2, 1.6
- (d) What is the motivation behind making the step value adaptive in a DM system? Explain. How such adaptation can be made. (10+4)
4. (a) (i) Give an example where Time Division Multiplexing (TDM) is required. (5+12)
- (ii) A DM1/2 multiplexer combines four DS1 signals, each of rate 1.544 Mbps and generates a time multiplexed signal of rate 6.312 Mbps. The frame format for such a DM1/2 multiplexer is given in Figure 4(a). Now explain why such framing is required. Particularly, describe the functions of the overhead bits: M, C and I's.
- | | | | | | | | | | | | |
|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|
| M_0 | [48] | C_A | [48] | F_0 | [48] | C_A | [48] | C_A | [48] | F_1 | [48] |
| M_1 | [48] | C_B | [48] | F_0 | [48] | C_B | [48] | C_B | [48] | F_1 | [48] |
| M_1 | [48] | C_C | [48] | F_0 | [48] | C_C | [48] | C_C | [48] | F_1 | [48] |
| M_1 | [48] | C_D | [48] | F_0 | [48] | C_D | [48] | C_D | [48] | F_1 | [48] |
- Figure: 4(a)
- (b) When does 'Inter-Symbol Interference (ISI)' occur in a digital transmission system? Explain how ISI can be eliminated using pulse shaping. (6+12)

SECTION - B

There are **FOUR** questions in this section. Answer any **THREE**.

You must give your answer to the point avoiding unnecessary writings.

Try to draw as many figures as possible and give explanation with block diagrams.

Symbols in questions have the usual meanings.

5. (a) Why do you need to modulate baseband signal? (3)
- (b) Show a comparison between DSB-SC, DSB+C (With Carrier), LSB-SC for the following (2+3+3+3+3)
- (1) Modulation for a tone of $m(t) = \cos 100t$
- (i) Draw basic spectrum of $m(t)$
 - (ii) Draw DSB-SC for $2m(t) \cos 100t$
 - (iii) Draw DSB+C for $2(1 + m(t))\cos 1000t$
 - (iv) Draw LSB-SC from figure of (ii) by suppressing USB
- (2) Bandwidth used for these modulation techniques.

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Contd ... Q. No. 5

- (c) Draw the signals obtained after encoding the bit stream '101011001' by each of the following digital line coding techniques: (i) Polar NRZ-1 (ii) Differential Manchester. **(6+12)**

Analyze these techniques in terms of the following characteristics.

- (i) Bandwidth requirement
- (ii) Synchronization capability
- (iii) DC value suppression
- (iv) Error detection capability

6. (a) Show the modulation procedure with block diagram, demodulation procedure with block diagram and bandwidth value only for the following.

- (i) PSK (Phase shift keying) **(3+3+1)**

- Modulation
- Coherent Demodulation
- Bandwidth

- (ii) FSK (Frequency shift keying) **(3+4+1)**

- Modulation
- Non-coherent Demodulation
- Bandwidth

- (iii) DPSK (Differential Phase shift keying) **(3+4+1)**

- Modulation
- Non-coherent Demodulation
- Bandwidth

For modulation case you must give an example binary sequence and draw its modulated signal (approximately).

For Demodulation case explain with either diagram of as few lines as possible.

- (b) Draw block diagram for a simple demodulation technique for FM (Frequency Modulation). **(4)**

- (c) Three signals, each covering the range 10-15 KHz is to be transmitted simultaneously using Frequency Division Multiplexing (FDM). The resultant signal should cover the range 80-95 KHz. Describe how to generate the desired frequency multiplexed signal. **(8)**

7. (a) (i) Explain the relationship between FM (Frequency Modulation) and MP (Phase Modulation) in term of angle and instantaneous frequency of carrier signal. **(4+7+2+2)**

(ii) Show that the bandwidth of FM signal is infinite.

(iii) Under what condition FM is called Narrowband FM (NBFM).

(iv) Applying the above condition on the above expression of an FM signal, show that a narrowband FM signal can be approximated as:

$$f_{NBFM}(t) \approx A[\cos\omega_c t - k_f a(t) \sin\omega_c t] \quad \text{where } a(t) = k_f \int_{-\infty}^t f(a) da$$

Contd ... P/4

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Contd ... Q. No. 7

- (b) Show with figure how phase continuation is maintained during MSK, what are the prerequisites for carrier frequency to work with MSK. (8+4)
- (c) On the process of recovering incoming carrier explain the following method briefly. (3+5)
- (i) Pilot carrier
 - (ii) Phase locked loop (PLL) (Draw Block diagram)
8. (a) Show that, for a time function $f(t)$, the signal, $f_{SSB}(t) = f(t)\cos\omega_c t + f_h(t)\sin\omega_c t$ represents a lower sideband SSB-SC signal where $f_h(t)$ represents Hilbert Transform of $f(t)$ and ω_c is the carrier frequency in rad/sec. (14)
- (b) Describe how Quadrature Amplitude Modulation (QAM) is performed. Explain why QAM is bandwidth efficient than DSB-SC modulation. (4+5)
- (c) For Vestigial Sideband (VSB), derive that output filter $H_0(\omega) = \frac{1}{H_i(\omega + \omega_c) + H_i(\omega - \omega_c)}$ is used to reform the actual transmission signal that is distorted by transmitter input filter $H_i(\omega)$. [Show only the necessary equation] (8)
- (d) Describe how 'Scrambling' solves the synchronization problem with Bipolar-AM1. (4)

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) (i) In Bipolar NRZ-L line coding , 1 is represented by 5V and 0 is represented by –5V during bit duration. Now, explain why Bipolar NRZ-L suffers from synchronization problems for both long run of 1's and 0's. (4+4+4+4+4)
- (ii) With proper explanation, give an example of a line coding technique that uses the same bandwidth as Bipolar NRZ-L but solves the synchronization problem for long run of 1's (though synchronization problem for long 0's still exists).
- (iii) With proper explanation, give an example of a line coding technique that solves the synchronization problems for both long run of 1's and 0's but requires twice the bandwidth of Bipolar NRZ-L.
- (iv) Describe how 'Block Coding' attempts to solve the synchronization problem with "NRZ-I". Does "Block Coding" affect the bandwidth of the resulting signal? Explain with an example.
- (b) Five signals, each covering the range 20-80 KHz are to be transmitted simultaneously using Frequency Division Multiplexing (FDM). The resultant signal should cover the range 300-700 KHz where frequency gap (guard band) of 25 KHz is provided between each pair of consecutive channels for interference reduction. See Figure 1(b) for an illustration. Now, describe how to generate the desired frequency multiplexed signal. (15)

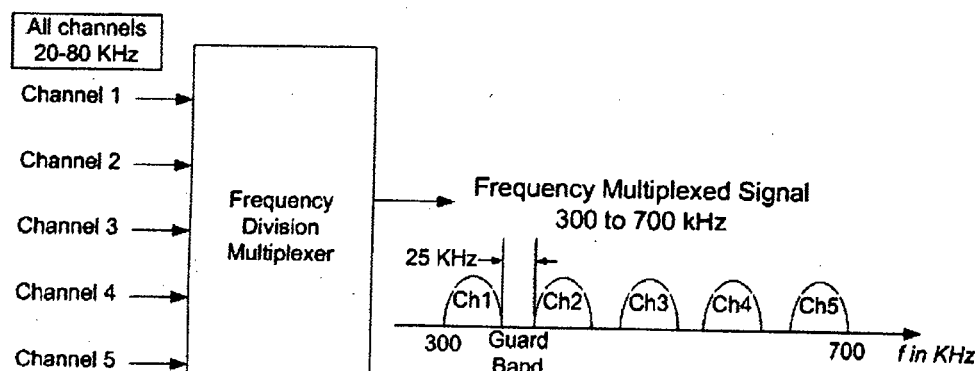


Figure 1(b)

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2. (a) For each the following three modulation techniques for binary digital data, determine the bandwidth of the modulated signal when the baseband bandwidth of the modulating signal is B Hz. (12)
- (i) Amplitude Shift Keying (ASK), (ii) Frequency Shift Keying (FSK), and (iii) Phase Shift Keying (PSK).
- (b) Draw the block diagrams for – (i) Noncoherent (asynchronous) detection of FSK, and (ii) coherent (synchronous) detection of FSK. (4+4)
- (c) (i) Explain with an example, how M-ary signaling achieves higher data rate than binary signaling using the same bandwidth but at the cost of more power. (7+8)
- (ii) Show that for M-ary FSK, the minimum frequency separation to ensure orthogonality among the FSK symbols is: $\delta f = 1/2T_b$ where T_b denotes the symbol duration.
3. (a) Explain how the circuit in Figure 3(a) functions as an envelope detector for AM (Amplitude Modulation) signal. Describe how to choose the constants 'R' and 'C' in the circuit for envelope detection of a given AM signal. (8+4)

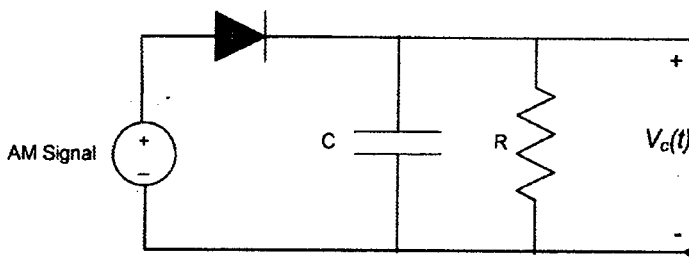


Figure 3(a)

- (b) Show that, for a real time function $f(t)$, the signal, $f_{SSB}(t) = f(t) \cos \omega_c t + f_h(t) \sin \omega_c t$ represents a lower sideband SSB-SC signal where $f_h(t)$ represents the Hilbert Transform of $f(t)$ and ω_c is the carrier frequency in rad/sec. (14)
- (c) Describe how Quadrature Amplitude Modulation (QAM) is performed. Explain why QAM is bandwidth efficient than DSB-SC modulation. (4+5)
4. (a) A baseband signal $f(t)$ of bandwidth B is modulating the frequency of a carrier at frequency ω_c (in rad/sec). Show that, the resulting frequency modulated signal can be expressed as – (5+2+8+5+3)

$$f_{FM}(t) = A \cos \left[\omega_c t + k_f \int_{-\infty}^t f(\alpha) d\alpha \right] \quad \text{where, } k_f \text{ is a constant.}$$

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Contd ... Q. No. 4(a)

Under what condition, an FM signal is called Narrowband FM (NBFM). Applying the condition on the above expression of an FM signal, show that a narrowband FM signal can be approximated as:

$$f_{NBFM}(t) \approx A \left[\cos \omega_c t - k_f a(t) \sin \omega_c t \right] \quad \text{where } a(t) = k_f \int_{-\infty}^t f(a) da$$

Draw the block diagram of a narrowband FM generator according to this expression.

What is the problem of generating NBFM signals in this way?

(b) 'Angle modulation is preferable over Amplitude modulation for systems where nonlinear amplifiers are used'. Justify the statement by showing how Angle Modulation is immune to nonlinearities whereas Amplitude Modulation suffers from distortion during nonlinear amplification.

(12)

SECTION - B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Define unit impulse function. Derive the sampling property of unit impulse function.

(8)

(b) The convolution of two functions $g(t)$ and $w(t)$, denoted by $g(t) * w(t)$, is defined by the

integral: $g(t) * w(t) = \int_{-\infty}^{\infty} g(\tau) w(t - \tau) d\tau$. Now, prove that the convolution of two signals in

the time domain becomes multiplication in the frequency domain and vice versa.

(14)

(c) The Fourier transform of a function $g(t)$ is expressed as: $G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$.

Find the Fourier transform of $g(t) = \Pi(t/\tau)$ in Figure 5(c).

(13)

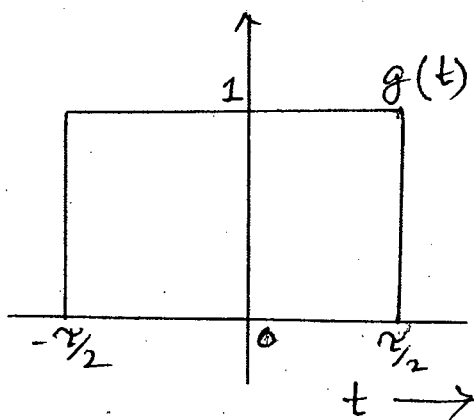


Figure: 5(c)

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6. (a) Describe how DPCM system works with suitable block diagrams of both transmitter and receiver. (12)
- (b) State and prove the Time-Shifting property and the Frequency Shifting property of Fourier Integral. (10)
- (c) How does 'Inter-Symbol Inference (ISI)' occur in digital transmission system? Explain how ISI can be eliminated using "pulse shaping". (13)
7. (a) Show that for a distortionless system, the amplitude response $|H(f)|$ must be a constant, and the phase response $\theta_h(f)$ must be a linear function of f going through the origin. (8)
- (b) Describe how TDM (Time Division Multiplexing) Digital Hierarchy works. Why bit stuffing and framing are used in TDM? (14)
- (c) A low-pass filter (Figure 7(c.1) transfer function $H(f)$ is given by (13)

$$H(f) = \begin{cases} (1 + k \cos 2\pi f T) e^{-j2\pi f t_d}, & |f| < B \\ 0, & |f| > B \end{cases}$$

A pulse $g(t)$, band-limited to B Hz (Figure 7(c.2), is applied at the input to this filter.

Show that the output is $y(t) = g(t - t_d) + \frac{k}{2} [g(t - t_d - T) + g(t - t_d + T)]$

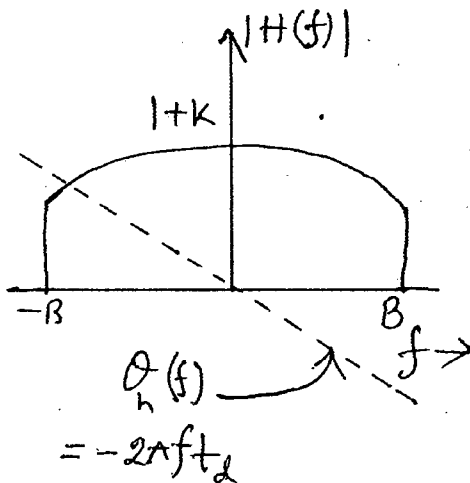


figure: 7(c.1)

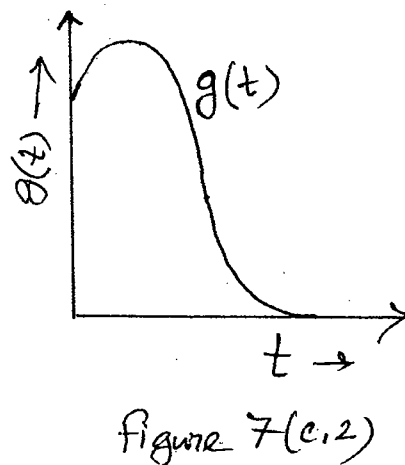


figure 7(c.2)

8. (a) Prove that the minimum sampling frequency for perfect signal recovery is $f_s = 2B$ Hz. Assume conventional meaning of the symbol. (15)
- (b) How quantization is done in a PCM system? Why non-uniform quantization is suitable for voice signal? How does companding achieve non-uniform quantization? (6+6+8=20)

Saifur
6/10/13

L-3/T-1/CSE

Date : 06/10/2013

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2011-2012

Sub : **CSE 311** (Data Communication I)

Full Marks : 210

Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION - A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) Draw the signals obtained after encoding the bit stream- '010100110' by each of the following two digital line coding techniques: (i) Polar NRZ-I and (ii) Bipolar-AMI.

Analyze these two techniques in terms of the following characteristics.

(5+10+5)

- (i) Bandwidth requirement
- (ii) Synchronization capability
- (iii) DC value suppression
- (iii) Error detection capability

Describe how 'Scrambling' solves the synchronization problem with Bipolar-AMI.

- (b)(i) Ten voice sources, each covering the frequency range from 20 Hz – 20 KHz are each to be transmitted using PCM technique as shown in Figure 1(b). Each signal is sampled at a rate of 44,000 samples/sec. Each sample is quantized and coded into 12 bits. The 10 PCM streams are then time-multiplexed into one composite PCM signal. (Neglect control and framing bits in the TDM frame that might be required). The composite PCM stream is fed to a 16-QAM modulator with carrier frequency, $f_c = 100$ MHz. Sinusoidal roll-off shaping with $r = 0.1$ is used. Find the transmission bandwidth at the output of the modulator.

(10)

- (ii) Consider again the problem in part (i) above. This time, a frame at the output of the time-division multiplexer is defined to be 1.25 ms long. 10 control bits are added per frame. What is the required transmission bandwidth at the output of the modulator?

(5)

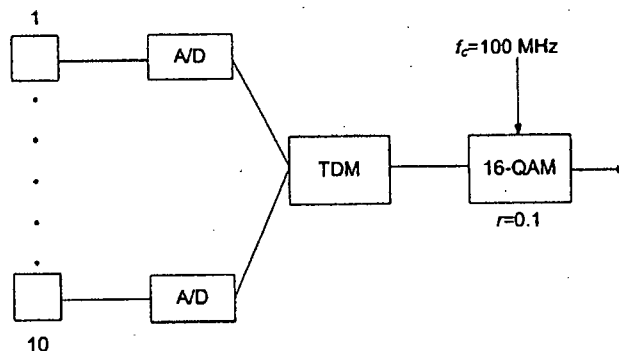
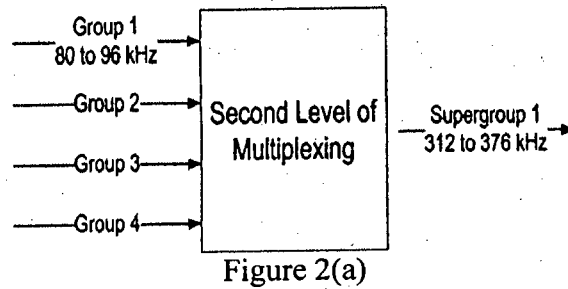


Figure 1(b)

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2. (a) Consider the second level of a two-level FDM hierarchy as shown in Figure 2(a). At this level, four groups (from level 1, not shown in the figure) each in the range 80 to 96 kHz are combined to generate a supergroup in the range 312 to 376 kHz using SSB (upper sideband) modulation. Explain the supergroup generation process using suitable diagram. Specify the frequency values that must be used at multiplier for each group/channel.

(10)



- (b) Explain how 'Frequency Modulation (FM)' and 'Phase Modulation (PM)' are related. Define narrowband FM and wideband FM. An FM signal can be demodulated by passing it through an ideal differentiator followed by an envelope detector. Explain why this method works.
3. (a) (i) One of the methods of coherent/synchronous detection of OOK signals is to multiply the incoming signal by the carrier frequency, as locally generated at the receiver, and then low pass-filtering the resultant multiplied signal. What happens when the local carrier is not exactly at the same frequency and same phase as the incoming carrier? Explain.
- (ii) There is another method for synchronously demodulating OOK signals that does not require exact frequency and phase synchronization. In this method, the incoming signal is sampled at the carrier frequency rate and then the sampled signal is passed through a low-pass-filter. Explain why this method works.
- (b) Draw the diagram of an envelope detector circuit. Why envelope detection cannot be used with PSK signals? Describe an asynchronous/non-coherent demodulation technique for PSK signals.
4. (a) Consider a 'Quaternary PSK (QPSK)' signal where the four phase angles are: $\pm\pi/4$, $\pm3\pi/4$. What is the maximum phase-shift at symbol transition point of the given QPSK signal? Describe how each of the following two techniques reduces the value of maximum phase-shift: (i) Offset QPSK and (ii) $\pi/4$ -shifted QPSK.
- (b) Describe how the following modulation techniques differ: (i) Normal AM, (ii) DSB-SC (Double-Sideband Suppressed-Carrier), (iii) SSB-SC (Single-Sideband Suppressed-Carrier) and (iv) VSB-SC (Vestigial-Sideband Suppressed-Carrier). Describe a scenario where Normal AM is required.

(8+5+12)

(12+8)

(3+3+9)

(3+8+8)

(8+8)

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SECTION - B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) The Fourier integral $F(\omega)$ of a non-periodic function $f(t)$ is defined as:

(8)

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Accordingly, find the Fourier integral of the function $f(t)$ shown in Figure 5(a).

- (b) Find the Fourier integral of function $g(t)$ shown in the Figure 5(b) from the Fourier integral of function $f(t)$ shown in Figure 5(a), and then the Fourier integral of function $h(t)$ shown in Figure 5(c) from the Fourier integral of function $g(t)$ using suitable properties of Fourier integral.

(8+6)

- (c) How is the bandwidth of a real time signal calculated? Accordingly which of the three functions $f(t)$, $g(t)$, and $h(t)$ shown in Figures 5(a), 5(b), and 5(c) respectively, has the lowest bandwidth? Explain your answer.

(5+8)

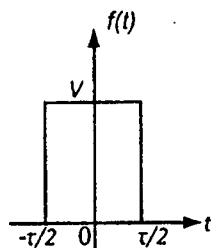


Figure 5(a)

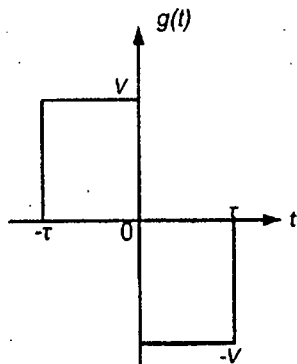


Figure 5(b)

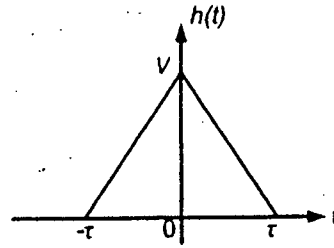


Figure 5(c)

6. (a) Explain the 'Linearity' and 'Time invariance' properties of a 'Linear Time Invariant (LTI)' system. Let $g(t)$ be output response of an LTI system with transfer function $H(\omega)$ to an input signal $f(t)$. Let $F(\omega)$ and $G(\omega)$ be the Fourier transform of $f(t)$ and $g(t)$ respectively. Show that, $G(\omega) = H(\omega)F(\omega)$.

(9+9)

- (b) An LTI system is called 'distortionless' when for any input function $f(t)$, the output of the system $g(t)$ is expressed as: $g(t) = Af(t - t_d)$, where A and t_d are constants. According to this definition, find out the transfer function $H(\omega)$ of a distortionless LTI system. Under what conditions, the RC circuit shown in Figure 6(b) can be approximated as a distortionless system.

(9+8)

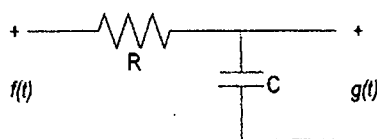
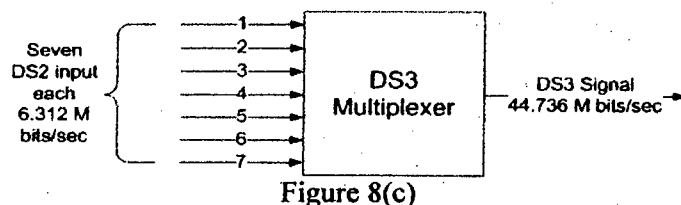


Figure 6(b)

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7. (a) In a PCM system, the input signal is limited to $\pm V$ volts. After sampling, the sampled signals are quantized into M uniformly spaced levels. For this system, show that, the mean power output SNR, $S_o/N_o = M^2 - 1$. (12)
- (b) In a PCM system, explain how quantization with non-uniform spacing of levels can be achieved without actually changing the uniform level spacing in the quantizer. (8)
- (c) Describe how 'Delta Modulation (DM)' system works with suitable block diagrams of both transmitter and receiver. (15)
8. (a) How does 'Inter-Symbol Interference (ISI)' occur in a digital transmission system? Explain how ISI can be eliminated using pulse shaping. (5+8)
- (b) Describe briefly how the following three issues are resolved in a Time Division Multiplexing (TDM) system. (i) Synchronization, (ii) Difference in the bit rates of the incoming channels, and (iii) Small variation in bit rate of an incoming channel. (9)
- (c) Consider a DS3 multiplexer as shown in Figure 8(c) that combines seven DS2 digital streams into one DS3 stream. Each incoming DS2 stream has a nominal bit rate of 6.312 Mbps. The bit rate of the output DS3 signal is 44.736 Mbps. Each DS3 frame contains 4704 data bits and 56 control bits. What is the maximum fractional increase over the nominal rate for each input channel that can be accommodated by the multiplexer? DS3 multiplexer allows one bit per frame to be stuffed per input signal channel. Is this sufficient to accommodate expected variation in the input rate? Explain your answer. (8+5)



SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) Consider the four signals in Figure 1(a), 1(b), 1(c) and 1(d) that are generated by encoding the bit-stream '0011100011' using four different digital line coding techniques. For each of these four line coding techniques identify and write the bit encoding rules (i.e., how 0 and 1 are represented). Compare the four techniques with respect to the following characteristics.

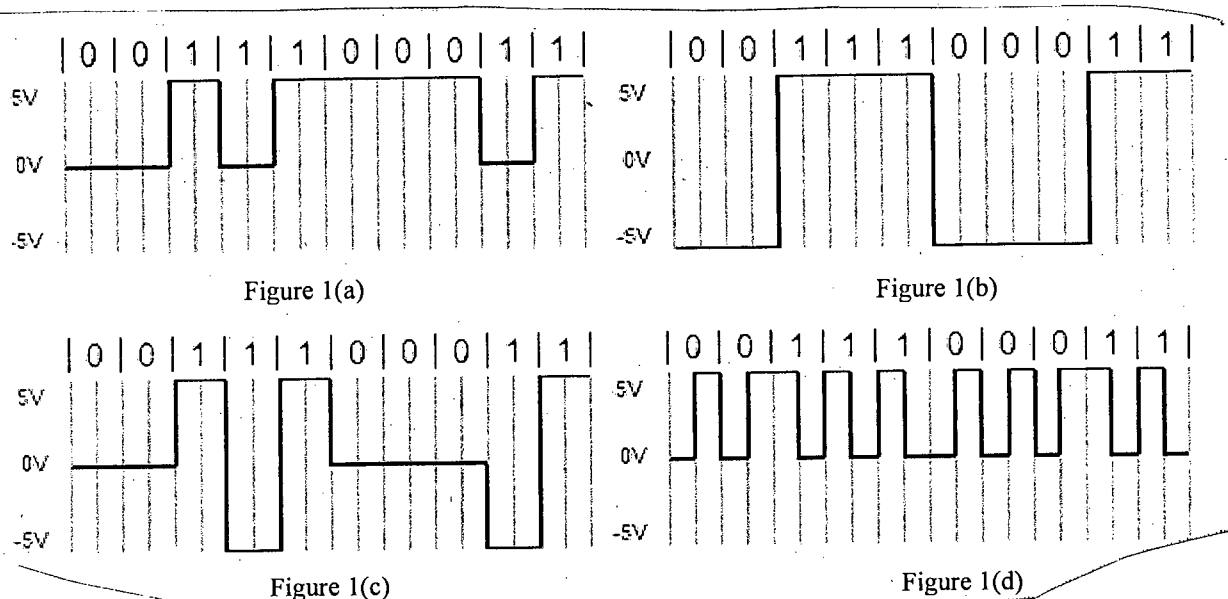
(4+16=20)

(i) Bandwidth requirement

(ii) Synchronization capability

(iii) DC value suppression

(iv) Error detection capability



- (b) Three signals, each covering the range 10-15 KHz are to be transmitted simultaneously using Frequency Division Multiplexing (FDM). The resultant signal should cover the range 80-95 KHz. Describe how to generate the desired frequency multiplexed signal.

(8)

- (c) FSK transmission is used to transmit 1200-bit/s digital signals over a telephone channel. The FSK signals are to fit into the range 500 to 2900 Hz. The carrier frequencies are 1200 and 2200 Hz respectively. Find the baseband bandwidth required for the binary signals. Assuming sinusoidal roll-off shaping, what roll-off factor is required?

(7)

2. (a) Explain with an example the difference between bit rate and pulse rate and thus show the advantage of using multisymbol signaling. Describe how multi-symbol signal can be generated using QPSK. Also draw the block diagram of a QPSK signal generator.

(8+8+4=20)

Contd P/2

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Contd ... Q. No. 2

(b) How do Frequency Modulation (FM) and phase modulation (PM) differ? Show that the bandwidth of narrowband FM signal is $2B$ Hz when baseband bandwidth of the modulating signal is B Hz. (5+10=15)

3. (a) Show that for a real time signal $f(t)$, the signal $f_{SSB}(t) = f(t)\cos\omega_c t + f_h(t)\sin\omega_c t$ represents a lower sideband SSB-SC signal where $f_h(t)$ represents the Hilbert Transform of $f(t)$. (15)

(b) There are two methods for generating SSB-SC signal: (i) Passing a DSB-SC signal through a band-pass filter; (ii) Using a circuit according to the expression for $f_{SSB}(t)$ as mentioned in Question 3(a). Now identify the practical difficulties involved in both the methods. Give example of a signal which is suitable for SSB-SC modulation using the first method. (8+5=13)

(c) Why Vestigial-sideband (VSB) signal is called so? Describe how VSB signal is generated. (3+4=7)

4. (a) What is the maximum number of pulses that can be transmitted over a bandwidth of B Hz? Explain your answer. (15)

(b) The amplitude spectrum for a sinusoidal roll-off filter is given as follows: (6+6+8=20)

$$\begin{aligned} |H(\Delta\omega)| &= \frac{1}{2} (1 - \sin(\pi/2)(\Delta\omega/\omega_x)) & |\Delta\omega| < \omega_x \\ &= 0 & |\Delta\omega| > \omega_x \\ &= 1 & -\omega_c < \Delta\omega < -\omega_x \end{aligned}$$

Sketch the amplitude spectrum. Does this filter satisfy the Nyquist Criterion of odd symmetry for generating wave-shape with zero intersymbol interference (ISI)? Explain your answer. Show that for the given sinusoidal roll-off filter $B = (1 + r)/2T$, where the symbols carry their usual meanings in the given context.

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) How is the bandwidth of a real time signal calculated? Let $f'(t)$ be the derivative of a real time signal $f(t)$. Which of $f(t)$ and $f'(t)$ has higher bandwidth? Explain your answer. (6+8=14)

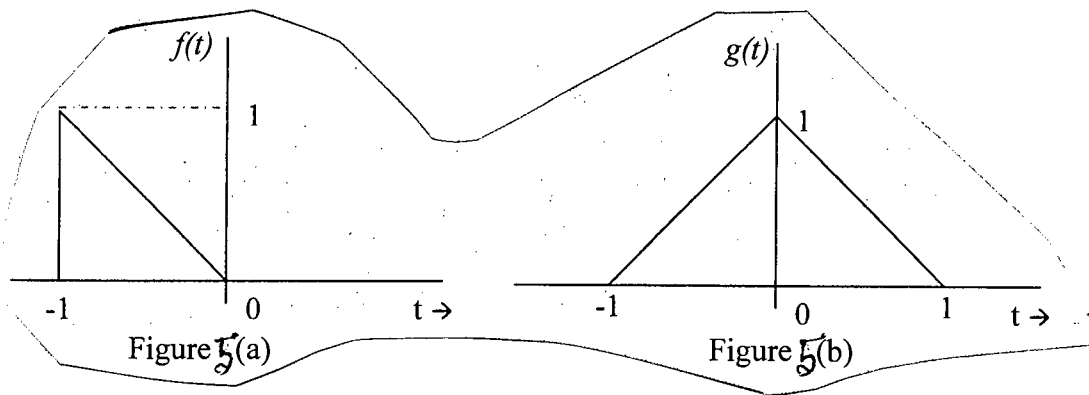
(b) The Fourier Transform of the triangular pulse $f(t)$ in Figure 5(a) is given as- (15+6=21)

$$F(\omega) = 1/\omega^2 (e^{j\omega} - j\omega e^{j\omega} - 1)$$

Using this information, and the time-shifting and time-scaling properties, find the Fourier Transforms of the signal $g(t)$ shown in Figure 5(b). Does the inverse-time bandwidth relation hold for this signal?

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Contd ... Q. No. 5(b)



6. (a) Find out and sketch the impulse response of an ideal low-pass filter with linear phase characteristic. Using the result explain why such an ideal filter is physically unrealizable.

(10+5=15)

(b) Let the signal $g(t)$ as shown in Figure 5(b) be input to a linear time invariant system with transfer function, $H(\omega) = j\omega$. Find out and sketch the output (in time domain) of the system.

(8)

(c) Let $f(t)$ be a real time signal band-limited to B Hz. Let $f_s(t)$ be the signal obtained by periodically sampling $f(t)$ at a rate of $2B$ samples per second. Show that if $f_s(t)$ is passed through an ideal low pass filter of bandwidth B Hz, then the output will be proportional to $f(t)$.

(12)

7. (a) How does a Quantizer with uniform spacing of levels work in a PCM system? When does quantization with non-uniform spacing result in better signal-to-quantization noise ratio than quantization with uniform spacing? Explain with an example.

(5+10=15)

(b) Explain how step size k' affects 'quantization noise' and 'overload noise' in a Delta Modulation (DM) system. How does adaptive delta modulation adjust k' to reduce both 'quantization noise' and 'overload noise'?

(14+6=20)

8. (a) What is Time Division Multiplexing (TDM)? What is the advantage of using TDM hierarchy? Explain why the output data rate of a multiplexer at any level in a TDM hierarchy is usually kept higher than the total of data rates of all its incoming channels.

(6+6+6=18)

(b) What is the advantage of using Differential Pulse Code Modulation (DPCM) system over Pulse Code Modulation (PCM) System? Discuss briefly on designing the predictor of a DPCM system.

(5+12=17)
